Kerala Polytechnic · SITTTR · Diploma EEE · Semester 3

3031

Analog & Digital Circuits

Complete Student Handbook · Program Core · 3 Credits · 60 Periods

4 Periods/Week | L:3 T:1 P:0 | 4 Modules | CO1–CO4 | 2 Series Tests

Module 1
Amplifiers & Oscillators

Module 2
Operational Amplifiers

Module 3
Number Systems, Logic & K-Map

Module 4
Combinational, Sequential & DAC/ADC

Handbook covers full theory · circuit diagrams · truth tables · K-Maps · solved problems · model QP · Malayalam concept hints

📋 Course Overview

Field	Details
Course Code	**3031**
Course Title	**Analog & Digital Circuits**
Programme	Diploma in Electrical & Electronics Engineering (EEE)
Semester	3
Credits	3
Course Category	Program Core
Periods per Week	4 (L:3, T:1, P:0)
Periods per Semester	60 (including 2 hrs Series Tests)
Prerequisites Course	Fundamentals of Electrical and Electronics Engineering (Sem 2)

This course builds a bridge between basic electronics and real-world electronic circuits. It covers both **analog circuits** (amplifiers, oscillators, op-amps) and **digital circuits** (logic gates, Boolean algebra, flip-flops, counters, DAC/ADC).

> 🔖 □ subject-□ analog circuits (amplifiers, oscillators, op-amps) □□ digital circuits (logic gates, flip-flops, counters) □□ □□□□□□□□□ □□□□□□□□. Electronics Engineering-□□□□ core topic □□□ □□□.

📖 How to Use This Handbook

1. **Start with Course Outcomes** — Understand what you need to achieve in each module before studying.
2. **Study module by module** — Each module has theory → diagrams → worked examples → quick check questions.

3. **Use the Formula Bank** — Before exams, review all formulas in one place. Practice derivations.
4. **Practice K-Map Bank** — K-Map simplification is a high-weight topic. Practice every example.
5. **Attempt Model QP** — Try the model question paper before reading answers. Check using the full answers provided.
6. **Tick Exam Checklist** — Use the final checklist to confirm all topics are covered before the exam.

> **Malayalam hints** appear below difficult concepts (marked 🔖). They are student-friendly explanations only — not definitions for exam writing.

🎯 Course Objectives

Objective 1: To understand the analog and digital electronic circuits.

This course gives students theoretical and conceptual understanding of how analog circuits (amplifiers, oscillators, operational amplifiers) and digital circuits (logic gates, combinational circuits, sequential circuits) work.

Objective 2: To use these circuits in electrical engineering applications.

Students will learn how to apply their circuit knowledge to real engineering problems — signal amplification, waveform generation, arithmetic operations, data storage, signal conversion (DAC/ADC), and more.

🔗 Course Prerequisites

Before studying this course, students must be familiar with the following topics from **Fundamentals of Electrical and Electronics Engineering (Semester 2)**:

#	Topic	Why Needed
1	Knowledge of electronics components (resistors, capacitors, diodes, transistors)	Circuit analysis and design
2	Transistor biasing & configurations (CE, CB, CC)	Needed for amplifier and oscillator circuits
3	Transistor operation as a switch	Needed for digital logic gate understanding

✅ Course Outcomes

On successful completion of this course, the student will be able to:

CO1 **Classify amplifiers and oscillators.**

Duration: 15 hours · Cognitive Level: Understanding

CO2 **Explain operational amplifiers and applications.**

Duration: 14 hours · Cognitive Level: Understanding

CO3 **Apply K-Map to simplify Boolean expressions.**

Duration: 14 hours · Cognitive Level: Applying

CO4 **Explain the various combinational, sequential and data conversion circuits.**

Duration: 15 hours · Cognitive Level: Understanding

Series Tests: 2 hours (1 hour after Module 2, 1 hour after Module 4)

🗺️ CO-PO Mapping

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	–	–	–	–	–	–
CO2	2	–	–	–	–	–	–
CO3	**3**	–	–	–	–	–	–
CO4	2	–	–	–	–	–	–

Mapping Scale: `3 = Strongly Mapped` `2 = Moderately Mapped` `1 = Weakly Mapped` `– = Not Mapped`

PO1 = Engineering Knowledge — the ability to apply foundational engineering principles. All COs in this course map to PO1 because the course directly builds circuit analysis and design knowledge required for an electronics/electrical engineer.

CO3 has the strongest mapping (3) because it involves higher-order applying skill (K-Map simplification), requiring active engineering knowledge application.

🛣️ Module-wise Learning Roadmap

Module 1 · 15 hrs

Amplifiers & Oscillators

- Multistage amplifiers
- Power amplifiers (A, B, C)
- Push-pull amplifiers
- Feedback (positive/negative)
- Oscillators & Barkhausen criterion
- RC phase shift, Crystal oscillator
- Astable & Bistable multivibrators

Module 2 · 14 hrs

Operational Amplifiers

- Ideal vs practical op-amp
- Virtual ground concept
- Inverting & non-inverting amp
- Summing amp, Integrator, Differentiator
- Comparators
- Precision rectifier

Series Test I after this

Module 3 · 14 hrs

Module 4 · 15 hrs

Digital Basics & K-Map

- Number systems & conversions
- Binary arithmetic & complements
- Logic gates (basic, universal, XOR)
- Boolean algebra & De Morgan's
- SOP / POS forms
- K-Map (up to 3 variables)

Combinational, Sequential & DAC/ADC

- Half adder, subtractor, full adder
- Multiplexer, Demultiplexer
- SR, D, JK, T flip-flops
- Shift registers (SISO/SIPO/PIPO/PISO)
- MOD-8/7/6 ripple counters
- DAC (Binary weighted, R-2R)
- ADC (Ramp type)

Series Test II after this

01

CO1 · 15 Hours · Understanding

Amplifiers and Oscillators

Multistage amplifiers · Power amplifiers · Feedback · Oscillators · Multivibrators

1.1 What is an Amplifier?

Amplifier

A device that increases the strength (voltage, current, or power) of an electrical signal without changing its shape. It uses a small input signal to control a larger output signal drawn from a DC power supply.

🔖 Amplifier = □□□□□□ signal-□□ □□□□□□□□□□□□□ device. Input signal □□□□□□□, output signal □□□□□□. Shape □□□□□□□□, strength □□□□□□.

VOLTAGE GAIN

```
Av = Vout / Vin
```

CURRENT GAIN

```
Ai = Iout / Iin
```

POWER GAIN

```
Ap = Pout / Pin = Av × Ai
```

Need for Multistage Amplification

A single transistor amplifier stage provides limited gain. For high gain requirements (e.g., radio receivers, audio systems), multiple amplifier stages are connected in series. The overall gain of a multistage amplifier equals the product of individual stage gains.

> 🔖 □□□□□□ transistor □□□□□□□□ gain □□□□□□. □□□□□□ □□ stages (multistage) □□□□□□□□□□□□□□.
> Total gain = stage 1 gain × stage 2 gain × ...

MULTISTAGE TOTAL GAIN

```
A_total = A1 × A2 × A3 × ... × An
```

1.2 Coupling Schemes in Multistage Amplifiers

> **Coupling**
>
> The method of connecting the output of one amplifier stage to the input of the next stage while blocking DC and passing AC signal.

Types of Coupling (listing):

1. **RC Coupling** (Resistance-Capacitance Coupling)

2. **Transformer Coupling**
3. **Direct Coupling** (DC Coupling)

Importance of Coupling:

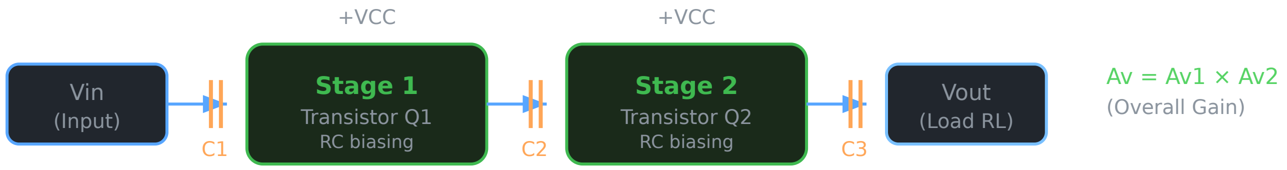
- Transfers AC signal from one stage to the next
- Blocks DC bias levels from one stage affecting the next
- Allows each stage to be independently biased

RC Coupled Two-Stage Transistor Amplifier

The most commonly used coupling method. Capacitors are used to couple stages — they pass AC but block DC. Each stage is independently biased.

> 🔖 RC coupling-□ capacitor □□□□□□□□□□□□□□□. Capacitor AC current-□□ □□□□□□□□□□□□, DC-□□ block □□□□□□□. □□□□□□□□□ □□□ stage-□□□□ DC bias next stage-□ effect □□□□□□□□□□□□□.

Two-Stage RC Coupled Transistor Amplifier — Block Diagram



1.3 Power Amplifiers

> **Power Amplifier**
>
> An amplifier designed to deliver large amounts of power to a load (e.g., speaker). Unlike voltage amplifiers, power amplifiers emphasise power output and efficiency over gain.

Characteristics of Power Amplifiers

- High power output capability
- Efficiency ($\eta = P_{out} / P_{DC} \times 100\%$)
- Low harmonic distortion
- Ability to handle large voltage and current swings

Comparison: Class A, Class B, and Class C Amplifiers

Parameter	Class A	Class B	Class C
Conduction Angle	360° (full cycle)	180° (half cycle)	<180° (less than half)
Quiescent Point (Q)	Centre of load line	At cut-off	Below cut-off
Efficiency	Max ~25–30% (resistive load), ~50% (transformer)	~78.5%	~80–100%
Distortion	Very low	Crossover distortion	High
Output Signal	Full cycle replica	Half cycle only	Highly clipped
Application	Audio pre-amplifiers	Audio power stages (push-pull)	RF transmitters, oscillators
Heat Dissipation	High (always ON)	Moderate	Low

🔖 Class A = full cycle amplify, efficiency □□□□□. Class B = half cycle, efficiency □□□□□□. Class C = less than half, efficiency □□□□□□□ □□□□□□. Distortion: A □□□□□□□ □□□□□, C □□□□□□□ □□□□□□.

1.4 Push-Pull Amplifiers

Push-pull amplifiers use two transistors — one for the positive half and one for the negative half of the input signal. The two outputs are combined to give a complete cycle. This eliminates even harmonic distortion and improves efficiency.

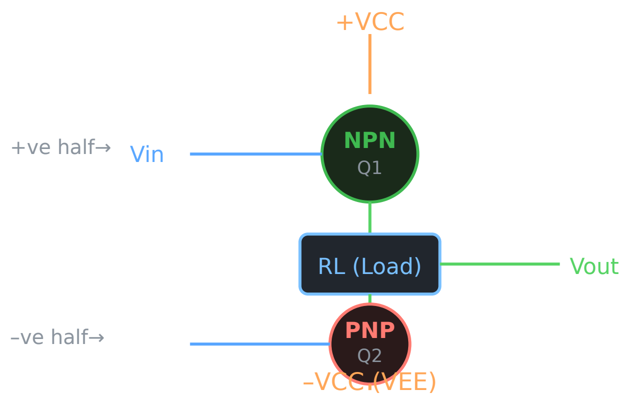
Complementary Symmetry Push-Pull Amplifier (Class B)

Uses one NPN and one PNP transistor (complementary pair). Both transistors operate in Class B — each conducts for exactly 180°. No biasing resistors are needed because the transistors are complementary.

- **NPN transistor** — conducts during positive half of input
- **PNP transistor** — conducts during negative half of input
- Combined output is a full sine wave
- Efficiency ≈ 78.5%

🔖 Push-pull-□ NPN transistor positive half amplify □□□□□□□□□, PNP transistor negative half □□□□□□□□□. □□ □□□□□□□□□ full sine wave □□□□□□□□□. Efficiency = 78.5%

Complementary Symmetry Class B Push-Pull Amplifier



1.5 Feedback in Amplifiers

Feedback

The process of returning a fraction of the output signal back to the input. Depending on the phase of the feedback signal relative to the input, feedback is classified as positive or negative.

Positive Feedback

- Feedback signal is **in phase** with the input signal
- Increases overall gain
- Can cause instability and oscillations
- **Used in:** Oscillator circuits

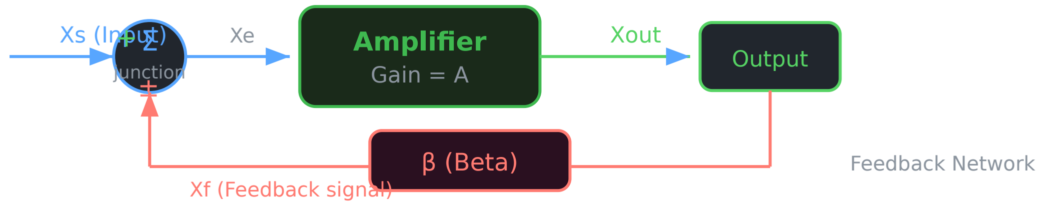
Negative Feedback

- Feedback signal is **out of phase** (180°) with the input signal
- Reduces overall gain
- Improves stability, bandwidth, reduces distortion
- **Used in:** Amplifier circuits

🔖 Positive feedback = output-□□□□ part input-□ add □□□□□□□ → gain □□□□□, oscillation □□□□□□□□. Negative feedback = output-□□□□ part input-□□ oppose □□□□□□□□□□ → gain □□□□□□□ stability □□□□□.

Feedback Block Diagram

Basic Feedback Amplifier Block Diagram



Key formulas (Feedback):

`Gain with feedback: Af = A / (1 ± Aβ)`

`Negative feedback: Af = A / (1 + Aβ) → gain reduced, stable`

`Positive feedback: Af = A / (1 − Aβ) → gain increased`

1.6 Oscillators

Oscillator

An electronic circuit that generates a continuous periodic waveform (AC output) without any external AC input signal. It converts DC power from the supply into AC output.

🔖 Oscillator = □□□ circuit □□□, □□□□□ external AC input □□□□□□ □□□□□. DC supply □□□□□□□□□ □□□ AC output □□□□□□□□□□. Radio waves generate □□□□□□□□□ clock signals □□□□□□□□□□□ □□□□□□□□□□□□□□.

Difference Between Amplifier and Oscillator

Amplifier	Oscillator
Requires external AC input	No external input required
Output = amplified version of input	Output = self-generated periodic waveform
External feedback not essential	Positive feedback is essential
Gain > 1	Net loop gain = 1 (at steady state)
Example: CE amplifier	Example: LC oscillator, RC oscillator

Basic Elements of an Oscillator Circuit

1. **Amplifier:** Provides required gain to sustain oscillations
2. **Feedback Network:** Returns a portion of output back to input (positive feedback)
3. **Frequency-Determining Element:** RC network, LC tank circuit, or crystal — sets oscillation frequency
4. **DC Power Supply:** Provides energy to the circuit

Barkhausen's Criterion for Oscillations

For an amplifier with positive feedback to oscillate, two conditions must be satisfied:

$$|A\beta| = 1 \qquad \text{(Loop Gain = 1)}$$

$$\angle A\beta = 0° \text{ or } 360° \qquad \text{(Phase Shift = 0° or 360°)}$$

- $|A\beta| = 1$ means the loop gain magnitude must be exactly 1 to sustain oscillations.
- If $|A\beta| < 1$, oscillations die out (damped). If $|A\beta| > 1$, oscillations grow (unstable). At $|A\beta| = 1$, oscillations are sustained.
- Total phase shift around the loop must be 0° or 360° (equivalent) for positive feedback.

🔖 Barkhausen criterion: oscillation □□□□□□□□□□□□□□ (1) loop gain = 1 □□□□□□□□□□, (2) total phase shift = 0° or 360° □□□□. □□□ □□□□□□□□□□□□□□ oscillator-□ output □□□□□ □□□□□□□□□□□□.

Classification of Oscillators

Based on output waveform:

- **Sinusoidal Oscillators** — produce pure sine wave output
- **Non-sinusoidal (Relaxation) Oscillators** — produce square, triangular, or sawtooth waves

Based on frequency-determining element:

- RC Oscillators (low-frequency audio range)
- LC Oscillators (radio frequency range)

- Crystal Oscillators (very stable, high frequency)

1.7 Sinusoidal Oscillators

Classification (listing):

- RC Phase Shift Oscillator
- Wein Bridge Oscillator
- Hartley Oscillator (LC type)
- Colpitts Oscillator (LC type)
- Crystal Oscillator

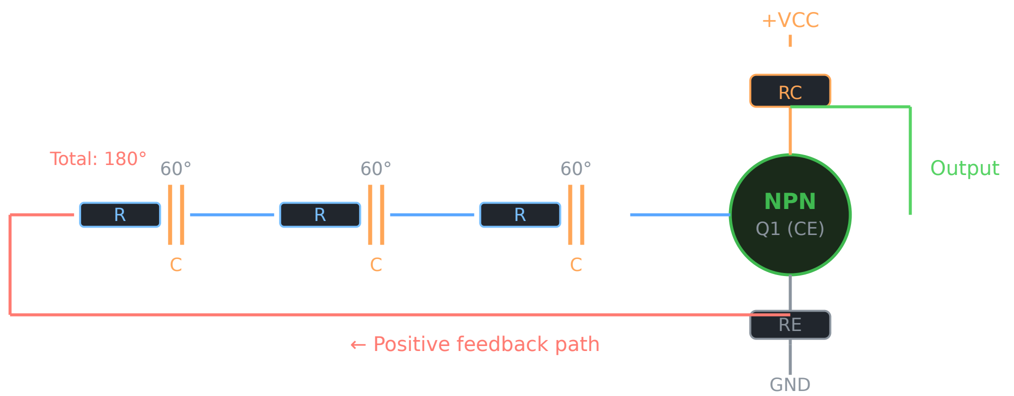
RC Phase Shift Oscillator

Uses a transistor amplifier and an RC feedback network. The RC network introduces a 180° phase shift. The transistor CE configuration introduces another 180° shift. Total = 360° → satisfies Barkhausen phase condition.

- Uses 3 RC sections each contributing 60° phase shift
- Total RC network phase shift = 3 × 60° = 180°
- Transistor CE amplifier phase shift = 180°
- Total loop phase = 360° ✓
- Frequency of oscillation: `f = 1 / (2π√6 RC)`
- Gain required: Av ≥ 29 (from analysis)

> 🔖 RC Phase shift oscillator: 3 RC sections + transistor amplifier. RC network 180° phase shift □□□□□□□□□□□□. Transistor CE 180° □□□□□□□□□□□□. Total = 360°. □□□ Barkhausen condition satisfy □□□□□□□□□□.

RC Phase Shift Oscillator — Circuit Diagram



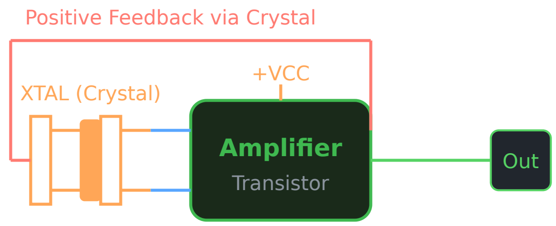
Crystal Oscillator

Uses a piezoelectric crystal (usually quartz) as the frequency-determining element. When mechanical stress is applied to the crystal, it generates an electrical voltage (piezoelectric effect). The crystal has a very precise natural resonant frequency.

- Extremely stable frequency output
- Very high Q-factor (10,000 to 100,000)
- Very low frequency drift with temperature
- Used in clocks, computers, communication equipment

> 🔖 Crystal oscillator = quartz crystal use □□□□□□□□□ oscillator. Frequency □□□□ stable □□□. Computer clock, mobile phone, GPS—□□□□□□□□ crystal oscillator use □□□□□□□□□□.

Crystal Oscillator — Basic Circuit



1.8 Non-Sinusoidal (Relaxation) Oscillators

Classification (listing):

- Multivibrators (Astable, Monostable, Bistable)
- Blocking Oscillators
- Sawtooth wave generators

Multivibrators

Multivibrators are switching circuits that produce square or rectangular wave outputs. They use transistors operating as switches (in saturation or cut-off).

Type	Stable States	Triggering	Output
Astable	0 (no stable state)	None (self-triggering)	Continuous square wave
Monostable	1 stable state	External trigger needed	One pulse per trigger
Bistable	2 stable states	External trigger for each transition	Stores 1 bit (flip-flop)

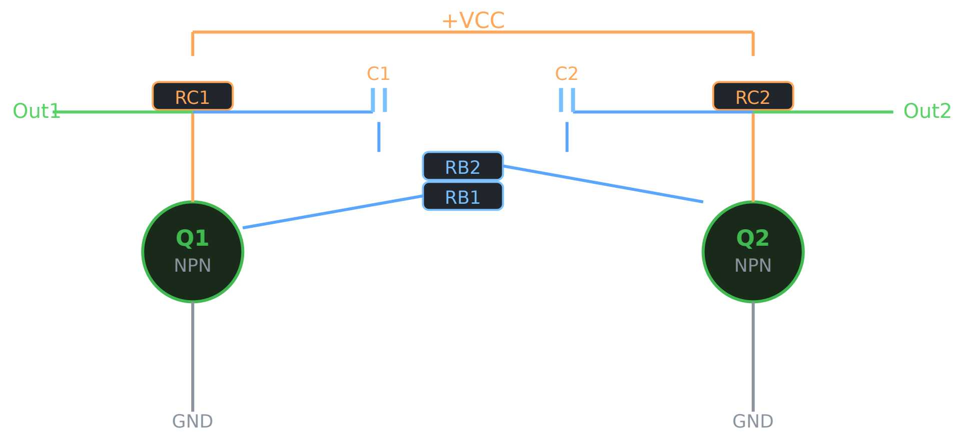
Astable Multivibrator

Has no stable state. Continuously switches between two quasi-stable states generating a square wave. Also called a **free-running multivibrator** or clock generator.

- Uses two transistors (Q1, Q2) cross-coupled via RC networks
- When Q1 is ON → Q2 is OFF, and vice versa
- Frequency determined by RC time constants

> 🔖 Astable multivibrator-□□ stable state □□□□□□□□□□. □□□ continuously Q1 ON/Q2 OFF → Q1 OFF/Q2 ON □□□ □□□□□□ switch □□□□□□□□□□□□□□□□□□. Output = square wave.

Astable Multivibrator — Circuit Diagram



Bistable Multivibrator

Has two stable states. Stays in one state until an external trigger is applied. Each trigger switches it to the other state. This is the fundamental building block of digital memory — also called a **flip-flop** or **latch**.

- Two transistors Q1 and Q2 cross-coupled
- State 1: Q1 ON, Q2 OFF
- State 2: Q1 OFF, Q2 ON
- External trigger required to change state

> 🔖 Bistable multivibrator = 2 stable states □□□□□. □□□ state-□ Q1 ON, Q2 OFF. Trigger signal □□□□□□□□□□□□ Q1 OFF, Q2 ON □□□□. □□□□□ flip-flop-□□□□ □□□□□□□□□.

Applications of Oscillators

- Clock generation in computers and microprocessors
- RF signal generation in radio transmitters
- Audio signal generation (audio oscillators)
- Waveform generation (function generators)
- Timing circuits and frequency synthesis

- Square wave generation (astable multivibrators)
- Pulse generation in digital systems

Module 1 — Quick Check Questions

1. What is Barkhausen's criterion? ▶

For an amplifier to oscillate, (1) Loop gain $|A\beta|$ must equal 1, and (2) total phase shift around the loop must be 0° or 360°. If gain > 1, oscillations grow; if < 1, they die out.

2. Compare Class A, B, C amplifiers. ▶

Class A: 360° conduction, low efficiency (~30%), least distortion. Class B: 180° conduction, efficiency ~78.5%, crossover distortion. Class C: < 180° conduction, highest efficiency (~80–100%), most distortion — used in RF applications.

3. What is the difference between positive and negative feedback? ▶

Positive feedback: output fed back in phase → gain increases → used in oscillators. Negative feedback: output fed back out of phase → gain decreases → used in amplifiers for stability.

4. What are the types of multivibrators? ▶

Astable (free-running, no stable state, generates square wave), Monostable (one stable state, generates pulse on trigger), Bistable (two stable states, flip-flop operation).

5. Why is crystal oscillator preferred over RC/LC oscillators? ▶

Crystal oscillators have very high Q-factor, giving extremely stable and precise frequency output. They have very low frequency drift with temperature variations. Hence used in computers, GPS, mobile phones.

02 Operational Amplifiers and Applications

CO2 · 14 Hours · Understanding · Series Test I after this module

Op-amp parameters · Inverting/Non-inverting · Summing · Integrator · Differentiator · Comparators · Precision Rectifier

2.1 Introduction to Operational Amplifier

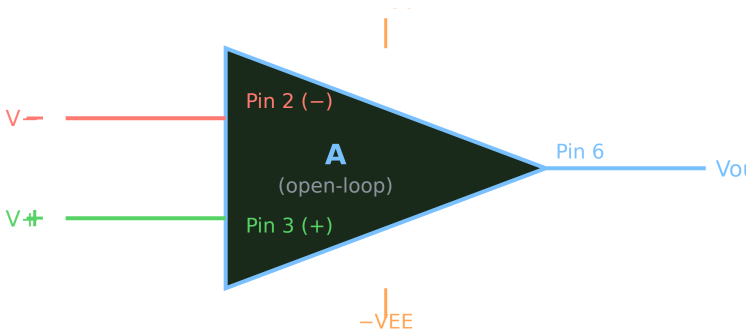
Operational Amplifier (Op-Amp)

A high-gain differential amplifier IC used to perform mathematical operations (addition, subtraction, integration, differentiation) on analog signals. The most popular IC op-amp is the **741**.

🔖 Op-amp = □□□ IC chip □□□. □□□ voltage amplify □□□□□□□, add □□□□□□□, subtract □□□□□□□, integrate □□□□□□□, differentiate □□□□□□□. Most popular: IC 741.

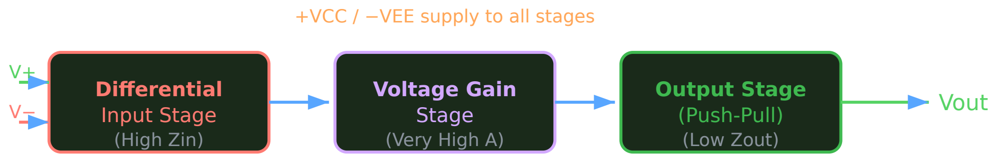
Op-Amp Symbol and Terminals

Op-Amp Symbol with Terminal Labels



Op-Amp Block Diagram

Internal Block Diagram of Op-Amp



2.2 Ideal vs Practical Op-Amp Parameters

Parameter	Ideal Op-Amp	Practical (IC 741)
Open-loop voltage gain (A)	∞ (infinity)	~2×10^5 (200,000)
Input impedance (Zin)	∞ (infinity)	~2 MΩ
Output impedance (Zout)	0 (zero)	~75 Ω
Bandwidth (BW)	∞ (all frequencies)	~1 MHz (at unity gain)
CMRR	∞	~90 dB
Slew Rate	∞	~0.5 V/μs
Offset Voltage	0	~2 mV
Supply voltage range	Any	±5V to ±18V

🔖 Ideal op-amp = gain, input impedance, bandwidth = infinity. Output impedance = 0. Practical (IC 741) = □□ finite values □□□. Exam-□ both compare □□□□□□□.

Key Parameter Definitions

CMRR (Common Mode Rejection Ratio)

The ability of the op-amp to reject signals that are common to both inputs. Higher CMRR means better noise rejection. CMRR = A_differential / A_common-mode. Expressed in dB.

Slew Rate

The maximum rate of change of output voltage per unit time. SR = dVout/dt (V/µs). Limits the highest frequency the op-amp can handle without distortion.

Virtual Ground

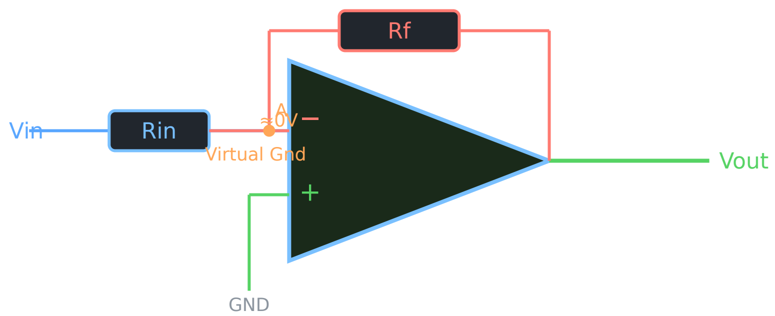
In an inverting op-amp circuit with negative feedback, the inverting input (V−) is approximately at 0V (ground potential) even though it is not physically connected to ground. This is because the very high gain forces V− ≈ V+ ≈ 0V.

🔖 Virtual ground: inverting input (−) practically 0V □□□, physically ground-□ connect □□□□□□□□□□□□□□□□□□□□. □□□ op-amp analysis-□ very useful □□□ — current calculation easy □□□□.

2.3 Inverting Amplifier

Input is applied to the inverting (−) terminal. Output is 180° out of phase with input. Negative feedback is applied via Rf from output to inverting input.

Inverting Amplifier Circuit



Derivation: Output Voltage of Inverting Amplifier

📐 Derivation — Inverting Amplifier Output Voltage

Step 1: Apply virtual ground concept → $V- = 0V$ (node A is at virtual ground)

Step 2: Current through Rin: $I_1 = (Vin - 0) / Rin = Vin / Rin$

Step 3: Since ideal op-amp draws no input current, all current flows through Rf: $I_1 = If$

Step 4: Voltage at output: $Vout = 0 - If \times Rf = -(Vin/Rin) \times Rf$

Result: `Vout = −(Rf / Rin) × Vin`

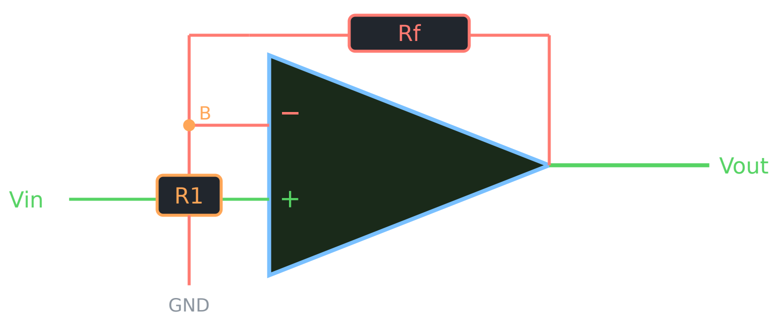
Voltage gain: $Av = Vout/Vin = -Rf/Rin$ (negative sign = 180° phase inversion)

🔖 Inverting amplifier output = −(Rf/Rin) × Vin. Negative sign → output □□ phase □□□□□□□□□□□□□□ (180° shifted). Rf □□□□□□□ □□□□□□□□□ gain □□□□□□.

2.4 Non-Inverting Amplifier

Input is applied to the non-inverting (+) terminal. Output is in phase with input. Negative feedback is via Rf and R1 (voltage divider).

Non-Inverting Amplifier Circuit



NON-INVERTING AMPLIFIER GAIN

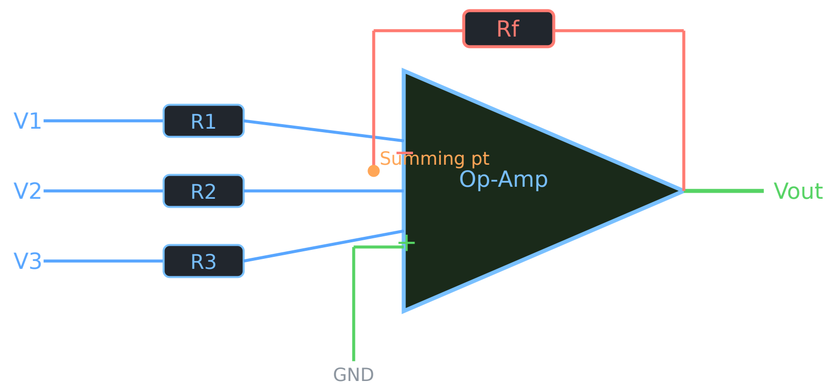
$$Av = 1 + (Rf / R1)$$

Output is in phase with input. Gain is always ≥ 1.

2.5 Summing Amplifier

An op-amp circuit that produces an output proportional to the algebraic sum of multiple input voltages. Multiple inputs are connected to the inverting terminal through separate resistors.

Summing (Adder) Amplifier Circuit



SUMMING AMPLIFIER OUTPUT

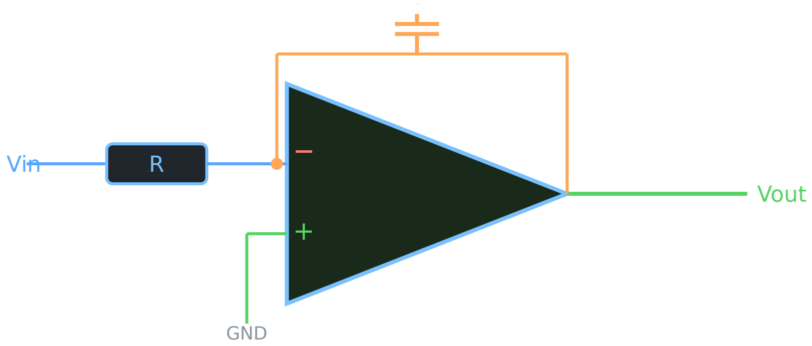
$$Vout = -Rf \times (V1/R1 + V2/R2 + V3/R3)$$

If R1=R2=R3=R: Vout = −(Rf/R)(V1+V2+V3)

2.6 Integrator

An op-amp circuit that performs mathematical integration of the input signal. A capacitor replaces Rf in the feedback path.

Integrator Circuit (Op-Amp)



📐 **Derivation — Integrator Output Equation**

Step 1: Virtual ground → $V- \approx 0V$

Step 2: Current through R: $I = Vin / R$

Step 3: This current charges capacitor C. Voltage across C = $(1/C)\int I\,dt$

Step 4: $Vout = -Vc = -(1/C)\int I\,dt = -(1/RC)\int Vin\,dt$

Result: $Vout = -(1/RC) \int Vin\,dt$

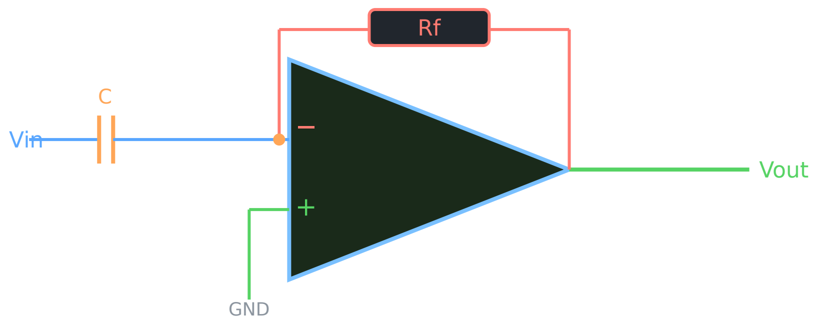
For a square wave input → integrator output is a **triangular wave**. For a sine wave input → output is cosine (phase shifted).

🔖 Integrator: capacitor feedback-□. Square wave input → triangular wave output. $Vout = -(1/RC)\int Vin\,dt$. RC value time constant control □□□□□□□□□□.

2.7 Differentiator

Performs mathematical differentiation. Capacitor is in the input path (replaces R), and R is in the feedback path.

Differentiator Circuit (Op-Amp)



📐 Derivation — Differentiator Output Equation

Step 1: Current through capacitor C: $I = C \times d(V_{in})/dt$

Step 2: Virtual ground → $V_- = 0$, so this current flows through Rf

Step 3: $V_{out} = -I \times R_f = -R_f \times C \times d(V_{in})/dt$

Result: $V_{out} = -RC \times dV_{in}/dt$

For a triangular wave input → differentiator output is a **square wave**. For a sine wave → output is cosine.

🔖 Differentiator: input-□ capacitor. Triangular wave → square wave output. $V_{out} = -RC \times dV_{in}/dt$. Input change rate □□□ output determine □□□□□□□□□□□.

2.8 Comparators

Comparator

An op-amp circuit (usually open-loop) that compares two voltages and produces a binary output — either +Vsat or −Vsat — depending on which input is larger.

🔖 Comparator: 2 voltages compare □□□□□□□□□ circuit. $V_+ > V_-$ □□□□□□□□ output = +Vsat. $V_+ < V_-$ □□□□□□□□ output = −Vsat. Binary (two-state) output.

Positive Voltage Level Detector

- Reference voltage Vref (positive) applied to non-inverting (+) terminal

- Input Vin applied to inverting (−) terminal
- When Vin < Vref → Vout = +Vsat
- When Vin > Vref → Vout = −Vsat
- Detects when input exceeds a positive reference level

Negative Voltage Level Detector

- Reference voltage Vref (negative) applied to inverting (−) terminal
- When Vin > Vref → Vout = +Vsat
- When Vin < Vref → Vout = −Vsat

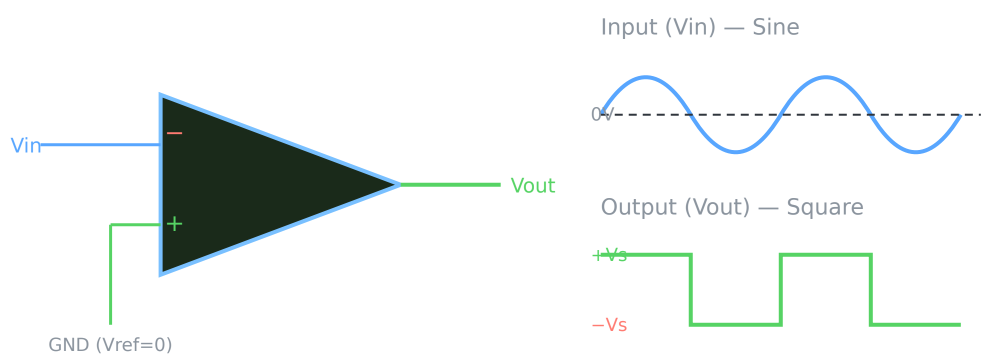
Zero Crossing Detector

A special case of comparator where Vref = 0V. The output switches polarity whenever the input crosses 0V.

- Non-inverting input connected to GND (Vref = 0)
- When Vin > 0 → Vout = −Vsat
- When Vin < 0 → Vout = +Vsat
- Converts a sine wave to a square wave

> 🔖 Zero crossing detector: Input 0V cross □□□□□□□□□□□ output switch □□□□. Sine wave input → Square wave output. Vref = 0V.

Zero Crossing Detector — Circuit and Waveforms



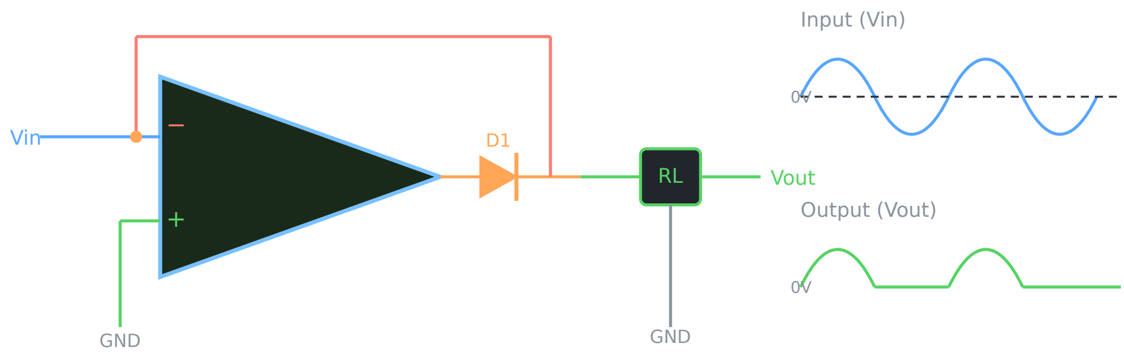
2.9 Precision Rectifier (Half-Wave)

A normal diode has a forward voltage drop of ~0.6–0.7V. For small signals (below 0.7V), a regular diode rectifier fails. A precision rectifier uses an op-amp to eliminate this drop — it can rectify even millivolt signals.

> 🔖 Normal diode rectifier-□□ 0.7V forward voltage drop □□□□□□. Small signals rectify □□□□□□□ □□□□□□□□. Precision rectifier-□ op-amp □□□□□□□□□□□□□□ — 0.7V drop □□□□□□□ exact rectification □□□□□□□.

- Diode D1 is in the feedback loop of op-amp
- During positive half: D1 conducts → output = input (no diode drop)
- During negative half: D1 reverse biased → op-amp output goes to −Vsat, Vout = 0

Half-Wave Precision Rectifier — Circuit and Waveforms



Module 2 — Quick Check Questions

1. What is virtual ground? ▶

In an inverting op-amp with negative feedback, the inverting input (V−) is held at approximately 0V by the feedback, even though it is not connected to ground. This is virtual ground. It simplifies circuit analysis.

2. Derive the gain of an inverting amplifier. ▶

Using virtual ground (V− = 0): Current through Rin = Vin/Rin. This current flows through Rf (op-amp draws no current). Vout = 0 − (Vin/Rin)×Rf = −(Rf/Rin)×Vin. Gain Av = −Rf/Rin.

3. What is the output of an integrator for a square wave input? ▶

A triangular wave. During the positive half of square wave, capacitor charges linearly → output is a negative ramp. During negative half → positive ramp. Combined result = triangular wave.

4. Why is a precision rectifier needed? ▶

A normal diode has ~0.7V forward bias drop. For signals below 0.7V, rectification fails. An op-amp precision rectifier eliminates this drop using the feedback loop, allowing accurate rectification of even millivolt-level signals.

5. Compare CMRR of ideal and practical op-amp. ▶

Ideal op-amp: CMRR = ∞ (perfect common-mode rejection). Practical IC 741: CMRR ≈ 90 dB. Higher CMRR means better ability to reject noise and interference that appears equally on both inputs.

CO3 · 14 Hours · Applying (K-Map)

03 Number Systems, Logic Gates, Boolean Algebra & K-Map

Binary · Hex · BCD · Arithmetic · Gates · Boolean Laws · K-Map simplification

3.1 Number Systems

Number System

A mathematical notation for representing numbers using a set of digits and a base (radix). Digital electronics uses binary (base-2) as its fundamental language because electronic circuits have two states: ON (1) and OFF (0).

🔖 Digital circuits-□ 0 (OFF) □□ 1 (ON) □□ □□□□□□□ □□□□□. □□□□□□ binary (base 2) number system use □□□□□□□□□□. Decimal (base 10) = □□□□□ daily life-□ □□□□□□□□□□□□□□□.

System	Base	Digits Used	Example
Decimal	10	0–9	245
Binary	2	0, 1	11110101
Hexadecimal	16	0–9, A–F	F5
Octal	8	0–7	305

Positional Weight Formulas

BINARY VALUE

$$N = b_n \times 2^n + \ldots + b_2 \times 2^2 + b_1 \times 2^1 + b_0 \times 2^0$$

HEXADECIMAL VALUE

$$N = h_n \times 16^n + \ldots + h_1 \times 16^1 + h_0 \times 16^0$$

3.2 Number Conversions

Binary to Decimal

🔢 **Example: Convert $(1101)_2$ to Decimal**

$= 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 1\times8 + 1\times4 + 0\times2 + 1\times1$

$= 8 + 4 + 0 + 1 =$ **13**

Decimal to Binary (Repeated Division by 2)

🔢 Example: Convert $(13)_{10}$ to Binary

13 ÷ 2 = 6, remainder **1** (LSB)

6 ÷ 2 = 3, remainder **0**

3 ÷ 2 = 1, remainder **1**

1 ÷ 2 = 0, remainder **1** (MSB)

Read remainders from bottom to top: **$(1101)_2$**

Hexadecimal to Decimal

🔢 Example: Convert $(2A)_{16}$ to Decimal

A = 10 (hexadecimal digit values: A=10, B=11, C=12, D=13, E=14, F=15)

$= 2 \times 16^1 + 10 \times 16^0$

= 32 + 10 = **42**

Decimal to Hexadecimal (Repeated Division by 16)

🔢 Example: Convert $(255)_{10}$ to Hex

255 ÷ 16 = 15, remainder **15 = F** (LSB)

15 ÷ 16 = 0, remainder **15 = F** (MSB)

Read from bottom: **$(FF)_{16}$**

Binary Coded Decimal (BCD)

Each decimal digit is represented separately using 4 binary bits.

Decimal	BCD (4 bits)	Decimal	BCD (4 bits)
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

Example: Convert $(29)_{10}$ to BCD

2 → 0010   9 → 1001

BCD of 29 = **0010 1001**

🔖 BCD = □□□ decimal digit-□□ separately 4-bit binary-□ represent □□□□□□□□□□. 29 = 2 (0010) 9 (1001) = 00101001 in BCD.

3.3 Binary Arithmetic

Binary Addition Rules

```
0 + 0 = 0
```

```
0 + 1 = 1
```

```
1 + 0 = 1
```

```
1 + 1 = 10 (0 with carry 1)
```

```
1 + 1 + 1 = 11 (1 with carry 1)
```

🔢 **Add: $(1011)_2 + (0110)_2$**

```
  1011
+ 0110
------
 10001   →  (17)₁₀ ✓ (11+6=17)
```

Binary Subtraction Rules

```
0 − 0 = 0
```

```
1 − 0 = 1
```

```
1 − 1 = 0
```

```
0 − 1 = 1 (borrow 1 from next higher bit)
```

Binary Multiplication Rules

```
0 × 0 = 0
```

```
0 × 1 = 0
```

```
1 × 0 = 0
```

```
1 × 1 = 1
```

🔢 Multiply: $(101)_2 \times (011)_2$

```
  101
× 011
-----
  101   (101 × 1)
 1010   (101 × 1, shifted left 1)
00000   (101 × 0, shifted left 2)
-----
01111  → (15)₁₀ ✓ (5×3=15)
```

3.4 Negative Number Representation: Complements

One's Complement

Invert all bits: 0 becomes 1, 1 becomes 0.

Example: One's complement of $(1010)_2$

Invert all bits: 1→0, 0→1, 1→0, 0→1

One's complement = **0101**

Two's Complement

TWO'S COMPLEMENT FORMULA

Two's complement = One's complement + 1

Example: Two's complement of $(1010)_2$

One's complement of 1010 = 0101

Add 1: 0101 + 0001 = 0110

Two's complement = **0110**

Binary Subtraction Using Two's Complement

🔢 **Subtract: $(1010)_2 - (0110)_2$ using two's complement**

Step 1: Find two's complement of subtrahend (0110): One's comp = 1001, + 1 = 1010

Step 2: Add: 1010 + 1010 = 10100

Step 3: Discard carry (MSB beyond 4 bits): 0100

Result = **0100 = $(4)_{10}$** ✓ (10−6=4)

> 🔖 Two's complement subtraction: subtrahend-□□□□ two's complement □□□□□□□□□□□□□, □□□□□□ add □□□□□□. Carry discard □□□□□□. Result = difference.

3.5 Logic Gates

Logic Gate

A basic electronic circuit that performs a logical operation on one or more binary inputs to produce a single binary output. Building blocks of all digital circuits.

AND Gate

Operation: Output is 1 only when ALL inputs are 1.

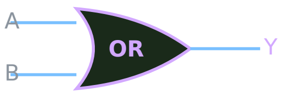
Boolean Expression: `Y = A · B` or `Y = AB`

A	B	Y=AB
0	0	0
0	1	0
1	0	0
1	1	**1**

OR Gate

Operation: Output is 1 when ANY input is 1.

Boolean Expression: `Y = A + B`

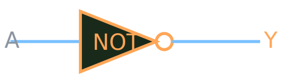


A	B	Y=A+B
0	0	0
0	1	**1**
1	0	**1**
1	1	**1**

NOT Gate (Inverter)

Operation: Output is complement of input.

Boolean Expression: `Y = A'` or `Y = Ā`



A	Y=A'
0	**1**
1	0

NAND Gate (Universal Gate)

Operation: NOT-AND. Output is 0 only when ALL inputs are 1.

Boolean Expression: `Y = (AB)'`

Universal gate: can implement any logic function using only NAND gates.

A	B	Y=(AB)'
0	0	**1**
0	1	**1**
1	0	**1**
1	1	0

NOR Gate (Universal Gate)

Operation: NOT-OR. Output is 1 only when ALL inputs are 0.

Boolean Expression: `Y = (A+B)'`

Universal gate: any logic function can be built using only NOR gates.

A	B	Y=(A+B)'
0	0	**1**
0	1	0
1	0	0
1	1	0

XOR Gate (Exclusive OR)

Operation: Output is 1 when inputs are **different**.

Boolean Expression: `Y = A ⊕ B = A'B + AB'`

Used in half adder, parity checker, comparator circuits.

A	B	Y=A⊕B
0	0	0
0	1	**1**
1	0	**1**
1	1	0

🔖 XOR gate: inputs different → output 1. Inputs same → output 0. Half adder-□ sum calculate □□□□□□□ use □□□□□□□□□□. NAND and NOR = universal gates (□□ □□□□□ □□□, □□□□□□

gates build □□□□□□□).

3.6 Boolean Algebra

Laws and Rules of Boolean Algebra (statements):

Law / Rule	AND Form	OR Form
Identity Law	$A \cdot 1 = A$	$A + 0 = A$
Null (Annulment)	$A \cdot 0 = 0$	$A + 1 = 1$
Idempotent	$A \cdot A = A$	$A + A = A$
Complement	$A \cdot A' = 0$	$A + A' = 1$
Double Complement	$(A')' = A$	
Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$
Associative	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$	$(A+B)+C = A+(B+C)$
Distributive	$A \cdot (B+C) = AB+AC$	$A+(B \cdot C) = (A+B) \cdot (A+C)$
Absorption	$A \cdot (A+B) = A$	$A+AB = A$

De Morgan's Theorems

Theorem 1: The complement of a product is the sum of complements.

$$(A \cdot B)' = A' + B'$$

Theorem 2: The complement of a sum is the product of complements.

$$(A + B)' = A' \cdot B'$$

De Morgan's is used to: break a NAND/NOR expression, simplify complex Boolean expressions, convert between SOP and POS forms.

🔖 De Morgan's: AND-□□□□ complement = OR of complements. OR-□□□□ complement = AND of complements. NAND ↔ OR with inverters. NOR ↔ AND with inverters.

3.7 Standard Forms: SOP and POS

SOP (Sum of Products)

A Boolean expression where product terms (ANDs) are summed (ORed). Each product term contains all variables in either true or complemented form. Also called minterm form. Example: Y = A'B + AB' + AB

POS (Product of Sums)

A Boolean expression where sum terms (ORs) are multiplied (ANDed). Each sum term contains all variables. Also called maxterm form. Example: Y = (A+B)(A+B')(A'+B)

Feature	SOP	POS
Structure	Sum (OR) of AND terms	Product (AND) of OR terms
Each term	Minterm	Maxterm
Output = 1 when	Any product term = 1	All sum terms = 1
K-Map representation	Group 1s	Group 0s
Implementation	AND-OR circuit	OR-AND circuit

3.8 Karnaugh Map (K-Map)

K-Map (Karnaugh Map)

A graphical tool to simplify Boolean expressions without using Boolean algebra. It arranges truth table values in a grid so adjacent cells differ by only one variable (Gray code order). Groups of 1s in the grid represent simplified product terms.

🔖 K-Map = Boolean expression simplify □□□□□□□□□□ chart method. Truth table values grid-□ arrange □□□□□□□□□□. Adjacent (neighbouring) cells 1-bit □□□□□□ differ □□□□□□□. 1s group □□□□□□ simplified expression □□□□□□□□□□□□□□□.

2-Variable K-Map Layout

Cell numbering (minterms):

	B=0	B=1
A=0	m0	m1
A=1	m2	m3

Minterm positions:

m	A	B
m0	0	0
m1	0	1
m2	1	0
m3	1	1

3-Variable K-Map Layout

Variables: A, BC (BC in Gray code: 00, 01, 11, 10)

A \ BC	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

⚠️ **Note:** BC columns follow Gray code order: 00 → 01 → 11 → 10 (NOT 00→01→10→11). This ensures adjacent cells differ by only one bit.

🔖 3-variable K-Map: BC column order = 00, 01, 11, 10 (Gray code). □□□□ adjacent cells □□□□□□ bit □□□□□□ change □□□□. □□□□□ K-Map-□□□□ key principle.

Grouping Rules in K-Map

- **Group sizes** must be powers of 2: `1` `2 (pair)` `4 (quad)` `8 (octet)`
- Groups can wrap around edges (left↔right, top↔bottom)
- Make groups as **large as possible** to maximise simplification
- Every 1 must be covered by at least one group
- Groups may overlap — share cells if it helps
- Fewer, larger groups = simpler final expression

Group Size	Name	Variables Eliminated
1 cell	Singleton	0 variables eliminated
2 cells	Pair	1 variable eliminated
4 cells	Quad	2 variables eliminated
8 cells	Octet	3 variables eliminated

K-Map SOP Simplification — Solved Example 1 (2-variable)

🗺️ Simplify: Y = Σm(1, 2, 3)

Step 1: Fill K-Map — place 1s at m1, m2, m3

A\B	0	1
0	0 (m0)	1 (m1)
1	1 (m2)	1 (m3)

Step 2: Group 1 (orange): m2 + m3 → both have A=1, B changes → A is constant → **A**

Step 3: Group 2 (blue): m1 + m3 → both have B=1, A changes → B is constant → **B**

Simplified SOP: **Y = A + B**

K-Map SOP Simplification — Solved Example 2 (3-variable)

🗺️ **Simplify: Y = Σm(0, 1, 2, 4, 5)**

Step 1: Fill 3-variable K-Map

A\BC	00	01	11	10
0	1(m0)	1(m1)	0(m3)	1(m2)
1	1(m4)	1(m5)	0(m7)	0(m6)

Step 2 — Group 1 (orange, 4 cells): m0,m1,m4,m5 → A varies, B varies, C=0 in all → **C'**

Step 3 — Group 2 (blue, 2 cells): m0,m2 → A=0, C varies, B=0 in both... check: m0(A=0,B=0,C=0), m2(A=0,B=1,C=0) → A=0 constant, C=0 constant, B varies → **A'C'**

Group 1 already covers m0 — no need to include it separately. Essential prime implicants: C' covers m0,m1,m4,m5. m2 needs A'C' (or absorbed into C' by noting m2 is also BC=10, C=0... covered by C').

Simplified SOP: **Y = C'** (m0,m1,m2,m4,m5 all have C=0 — quad + wrap confirms C')

Common K-Map Mistakes

- ❌ Using non-power-of-2 group sizes (e.g., 3 cells or 5 cells)
- ❌ Not using Gray code ordering for K-Map columns
- ❌ Not finding the largest possible groups
- ❌ Missing wrap-around groups (edges connect)
- ❌ Grouping 0s instead of 1s for SOP simplification
- ✅ Always double-check: every 1 must be in at least one group

Module 3 — Quick Check Questions

1. Convert $(1111\ 0101)_2$ to decimal. ▶

= 1×128 + 1×64 + 1×32 + 1×16 + 0×8 + 1×4 + 0×2 + 1×1 = 128+64+32+16+4+1 = **245**

2. Find two's complement of 10110. ▶

One's complement of 10110 = 01001. Add 1: 01001 + 1 = 01010. Two's complement = **01010**.

3. State De Morgan's theorems. ▶

Theorem 1: (A·B)' = A' + B' — complement of AND = OR of complements. Theorem 2: (A+B)' = A'·B' — complement of OR = AND of complements.

4. Why are NAND and NOR called universal gates? ▶

Any Boolean function (AND, OR, NOT, XOR, etc.) can be implemented using only NAND gates, or only NOR gates. They are called universal because they can replace all other gate types, reducing IC variety in circuit design.

5. What is the grouping rule in K-Map? ▶

Group sizes must be powers of 2 (1, 2, 4, 8). Groups must contain only 1s (for SOP). Larger groups give simpler expressions. Groups can wrap around edges. Every 1 must be covered. Aim for fewest, largest groups.

CO4 · 15 Hours · Understanding · Series Test II after this module

04

Combinational, Sequential and Data Conversion Circuits

Adders · MUX/DEMUX · Flip-flops · Shift Registers · Counters · DAC · ADC

4.1 Combinational vs Sequential Logic

Feature	Combinational Circuit	Sequential Circuit
Output depends on	Present inputs only	Present inputs + past state (memory)
Memory	No memory	Has memory (flip-flops)
Feedback	No feedback	Feedback present
Examples	Adder, MUX, Decoder	Flip-flop, Counter, Register
Clock signal	Not required	Usually required

🔖 Combinational circuit: output = only present inputs-□□ depend □□□□□□□□□□. Sequential: output = inputs + memory (past state). Flip-flop = sequential circuit-□□□□ basic building block.

4.2 Half Adder

Half Adder

A combinational circuit that adds two single-bit binary numbers (A and B) and produces a Sum and a Carry output. It cannot add a carry from a previous stage.

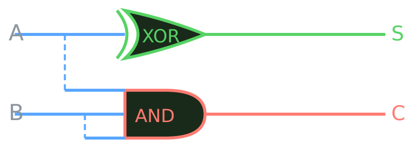
HALF ADDER BOOLEAN EXPRESSIONS

```
Sum (S) = A ⊕ B    (XOR gate)
```

```
Carry (C) = A · B    (AND gate)
```

A	B	Sum	Carry
0	0	0	0
0	1	**1**	0
1	0	**1**	0
1	1	0	**1**

Half Adder Logic Diagram



4.3 Half Subtractor

Half Subtractor

A combinational circuit that subtracts two single-bit binary numbers (A − B) and produces a Difference and a Borrow output.

HALF SUBTRACTOR BOOLEAN EXPRESSIONS

```
Difference (D) = A ⊕ B    (XOR gate)
```

```
Borrow (Br) = A' · B    (NOT A, then AND with B)
```

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

4.4 Full Adder

Full Adder

Adds three bits: A, B, and Carry-in (Cin). Produces Sum and Carry-out (Cout). Can be cascaded to add multi-bit numbers. Implemented using two Half Adders and one OR gate.

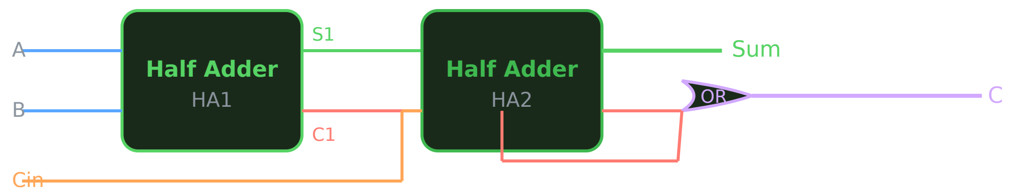
FULL ADDER EXPRESSIONS

```
Sum = A ⊕ B ⊕ Cin
```

```
Carry-out = AB + BCin + ACin
```

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full Adder Using Two Half Adders



🔖 Full adder = 2 half adders + 1 OR gate. 3 inputs (A, B, Cin) → Sum + Carry. Multi-bit adder = multiple full adders cascade □□□□□□□□□□.

4.5 Multiplexer (4×1) and Demultiplexer (1×4)

Multiplexer (MUX)

A data selector circuit. Selects one of many input data lines and routes it to a single output line, based on select input signals. Acts like a many-to-one switch.

Demultiplexer (DEMUX)

Opposite of MUX. Routes a single input data line to one of many output lines, based on select signals. Acts like a one-to-many switch.

🔖 MUX = many inputs → 1 output (select □□□ input output-□□□□□□). DEMUX = 1 input → many outputs (select □□□ output-□ data □□□□). TV remote channel select □□□□ □□□ MUX.

4×1 MUX — 4 data inputs (I0–I3), 2 select lines (S0, S1), 1 output Y

S1	S0	Output Y
0	0	I0
0	1	I1
1	0	I2
1	1	I3

1×4 DEMUX — 1 data input (D), 2 select lines (S0, S1), 4 outputs (Y0–Y3)

S1	S0	Active Output
0	0	Y0 = D
0	1	Y1 = D
1	0	Y2 = D
1	1	Y3 = D

4.6 Flip-Flops

Flip-Flop

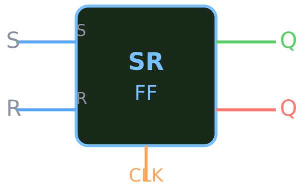
A bistable sequential circuit that stores one bit of information. Output changes on a clock edge (rising or falling). Building block of registers, counters, and memory circuits.

SR Flip-Flop

S=Set, **R**=Reset. S=1 → Q=1. R=1 → Q=0. S=R=1 → **forbidden state**.

S	R	Q(next)	State
0	0	Q (no change)	Hold
0	1	0	Reset
1	0	1	Set
1	1	X (invalid)	Forbidden

SR Flip-Flop Symbol

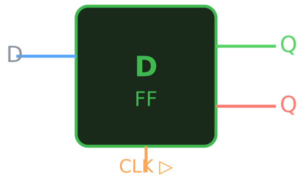


D Flip-Flop

D=Data. On clock edge, Q follows D. Eliminates forbidden state of SR FF. Used in registers.

D	Q(next)
0	0
1	1

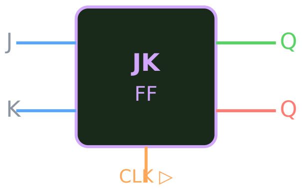
Q(next) = D



JK Flip-Flop

Improved SR FF. J=Set, K=Reset. J=K=1 → **toggle** (Q complements). No forbidden state.

J	K	Q(next)	Operation
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	Q'	Toggle

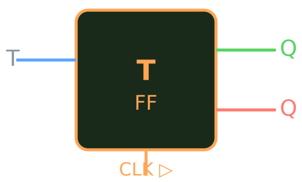


🔖 JK flip-flop: SR-□□□□ improved version. J=K=1 □□□□□□□□ output toggle (complement) □□□□□□□. Forbidden state □□□□. Counters-□ most used flip-flop.

T Flip-Flop

T=Toggle. When T=1, output toggles on every clock pulse. When T=0, holds state. Derived from JK FF with J=K=T.

T	Q(next)	Operation
0	Q	Hold
1	Q'	Toggle



Applications of Flip-Flops

- Data storage (registers)
- Counters and frequency dividers
- Shift registers
- Memory cells in RAM
- Synchronizers and pulse generators

4.7 Shift Registers

Shift Register

A sequential circuit made of flip-flops connected in series. Data bits can be shifted in or out serially or loaded/retrieved in parallel. Used for serial-to-parallel and parallel-to-serial data conversion.

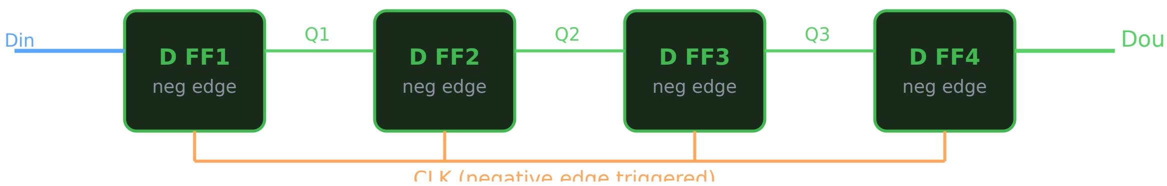
Classification of Shift Registers

Type	Full Form	Input	Output
SISO	Serial In Serial Out	Serial (1 bit at a time)	Serial (1 bit at a time)
SIPO	Serial In Parallel Out	Serial	Parallel (all bits at once)
PIPO	Parallel In Parallel Out	Parallel	Parallel
PISO	Parallel In Serial Out	Parallel	Serial

🔖 Serial = □□□ bit-□□ clock-□ □□□□□□□□□□□/□□□□□□□□□□. Parallel = □□□ □□□□ all bits. SIPO = data □□□ bit □□□ enter □□□□, □□□ time-□ all bits read □□□□□□□.

4-bit SISO Shift Register (Negative Edge Triggered D Flip-Flops)

4-bit SISO Shift Register using D Flip-Flops



Applications of Shift Registers

- Serial-to-parallel and parallel-to-serial data conversion
- Time delay circuits
- Ring counters and Johnson counters
- Data storage and data transfer in digital systems
- Universal shift registers in microprocessors

4.8 Counters

Counter

A sequential circuit that counts clock pulses and produces a binary sequence at its outputs. The MOD (modulus) of a counter = number of distinct states it goes through before resetting.

Classification of Counters

- **Asynchronous (Ripple) Counter** — flip-flops are clocked one after another; clock ripples through
- **Synchronous Counter** — all flip-flops clocked simultaneously
- **Up Counter** — counts upward (0,1,2,...)
- **Down Counter** — counts downward (...2,1,0)
- **Modulo-N Counter** — counts N states (MOD-N)

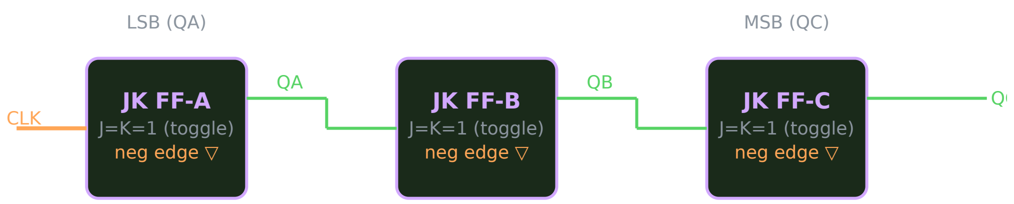
Feature	Asynchronous Counter	Synchronous Counter
Clock	FF1 clocked by input; each FF clocked by previous FF's output	All FFs clocked simultaneously
Speed	Slower (propagation delay accumulates)	Faster
Complexity	Simple	More complex
Glitch	More prone to glitches	Less prone

🔖 Asynchronous = ripple counter. □□□ FF-□□ previous FF-□□□□ output-□ □□□□□□ clock □□□□□□□□□□□. Synchronous = common clock. Faster, more complex.

MOD-8 Ripple Up Counter (3-bit, using negative edge JK FF)

Uses 3 negative-edge-triggered JK flip-flops with J=K=1 (toggle mode). Counts from 000 to 111 (0 to 7), then resets to 000. Total states = 2^3 = 8 = MOD-8.

MOD-8 Ripple Up Counter — Logic Diagram



MOD-8 Ripple Counter Sequence

Clock Pulse	QC	QB	QA	Decimal
0 (initial)	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7
8 (reset)	0	0	0	0 (repeats)

MOD-7 Ripple Up Counter

Uses 3 JK FFs (same as MOD-8) with a NAND gate feedback to reset when count reaches 7 (111). States: 0 to 6 (7 states). Counter resets to 000 when all three outputs are 1 simultaneously.

Reset condition for MOD-7: When QA=QB=QC=1 (count=7), feed into a NAND gate. Its output goes LOW → feeds into CLR (clear) input of all FFs → resets to 000 immediately. So 7 is a momentary state — effective count = 7 states (0–6).

MOD-6 Ripple Up Counter

Uses 3 JK FFs with NAND gate reset when count reaches 6 (QC=1, QB=1, QA=0 → actually when 110 appears). Reset condition: QB=QC=1 (count=6). States: 0 to 5 (6 states).

Reset condition for MOD-6: When QB=1 AND QC=1 (count=6=110), NAND gate resets all FFs to 000. Count cycles 0→1→2→3→4→5→0.

🔖 MOD-N counter: N states □□□□□□□□□. MOD-7 → 0 to 6. MOD-6 → 0 to 5. Extra states skip □□□□□□□ NAND gate feedback reset use □□□□□□□□□□□.

4.9 Data Conversion Circuits

Why Data Conversion?

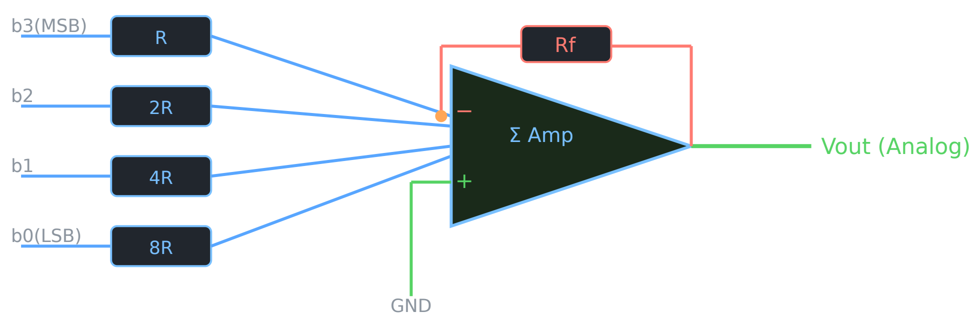
The real world is analog (temperature, pressure, sound). Digital systems process binary. Data converters bridge the gap:

- **DAC** (Digital-to-Analog Converter) — converts binary number to equivalent analog voltage
- **ADC** (Analog-to-Digital Converter) — converts analog signal to binary number

Binary Weighted DAC

Uses a resistor network where each resistor has a value proportional to the binary weight of the corresponding bit. Most significant bit (MSB) has smallest resistor (R), next bit has 2R, then 4R, etc. An op-amp summing amplifier combines the currents.

Binary Weighted DAC (4-bit)

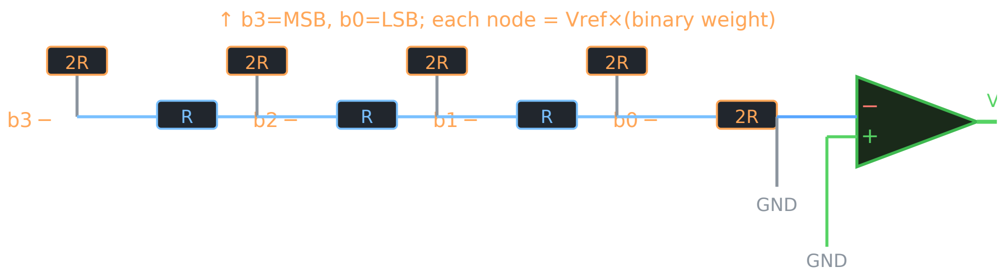


Disadvantage: Requires very precise resistors with wide range of values (R, 2R, 4R, 8R). For large bit numbers, higher resistors become impractical.

R-2R Ladder DAC

Uses only two resistor values: R and 2R, regardless of the number of bits. More practical and accurate than binary weighted DAC. Each node in the ladder divides the voltage by 2, equivalent to binary weighting.

R-2R Ladder DAC (4-bit) — Concept



Advantage: Only two resistor values (R and 2R) used for any number of bits — easier to fabricate in IC form with high precision.

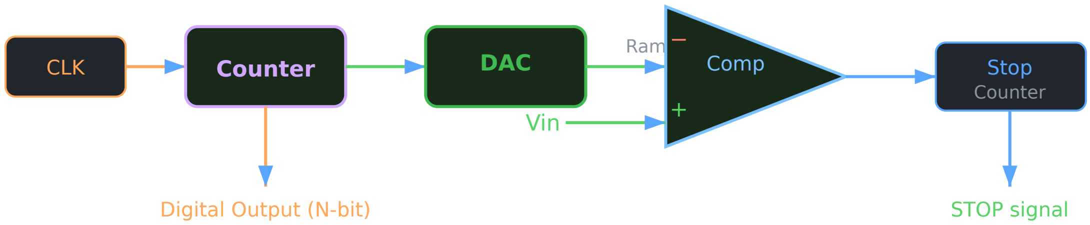
> 🔖 R-2R ladder DAC: R □□ 2R □□ □□□□□□ use □□□□□□□□□□. Binary weighted DAC-□ R, 2R, 4R, 8R... □□□□□□□□□□□□□. R-2R-□ precision better, manufacturing easy.

Ramp Type ADC

Converts analog input to digital output using a ramp (staircase) voltage and comparator. Also called staircase ADC or counter-type ADC.

- A counter starts from 0 and increments on each clock pulse
- Counter output is fed to a DAC which generates a staircase ramp voltage
- Ramp voltage is compared to Vin using a comparator
- When ramp voltage = Vin → comparator switches → counter stops → digital output = binary count

Ramp Type ADC — Block Diagram



> 🔖 Ramp ADC: Counter-□ □□□□□□ binary → DAC → ramp voltage. Ramp = Vin □□□□□□□□ comparator counter-□□ stop □□□□□□□□□□. Counter value = digital output. Simple but slow.

Module 4 — Quick Check Questions

1. What is a half adder? Give its Boolean expressions. ▶

A half adder adds two 1-bit numbers A and B. Sum = A⊕B (XOR). Carry = A·B (AND). Implemented using 1 XOR gate and 1 AND gate. Cannot add carry from previous stage.

2. What is the difference between combinational and sequential circuits? ▶

Combinational: output depends only on present inputs, no memory (e.g., adder, MUX). Sequential: output depends on present inputs AND past state (memory), uses flip-flops (e.g., counter, register).

3. What is the function of a JK flip-flop when J=K=1? ▶

When J=K=1, the JK flip-flop toggles — output Q complements on each clock pulse. This eliminates the forbidden state that exists in SR flip-flop when S=R=1.

4. How does a MOD-6 ripple counter differ from MOD-8? ▶

MOD-8 uses 3 JK FFs and counts from 0 to 7 (8 states) without reset logic. MOD-6 adds a NAND gate that resets the counter when count=6 (QB=QC=1), so it counts 0 to 5 (6 states only).

5. Why is R-2R ladder DAC preferred over binary weighted DAC? ▶

Binary weighted DAC needs resistors of values R, 2R, 4R, 8R... which become impractical for high bit numbers. R-2R ladder uses only two values (R and 2R) regardless of bit count, making it easier to manufacture with high precision in IC form.

∑ Formula Bank

All key formulas from all four modules in one place. Review before exams.

Module 1 — Amplifiers & Oscillators

Formula	Expression	Meaning
Voltage Gain	`Av = Vout / Vin`	Ratio of output to input voltage
Current Gain	`Ai = Iout / Iin`	Ratio of output to input current
Power Gain	`Ap = Pout / Pin = Av × Ai`	Ratio of output to input power
Multistage Gain	`A_total = A1 × A2 × ... × An`	Product of individual stage gains
Gain with Feedback	`Af = A / (1 ± Aβ)`	+ for negative, − for positive feedback
Barkhausen Criterion	`|Aβ| = 1 and ∠Aβ = 0° or 360°`	Condition for sustained oscillations
RC Phase Shift Frequency	`f = 1 / (2π√6 RC)`	Oscillation frequency of RC phase shift oscillator
Class B Efficiency	`η_max = π/4 ≈ 78.5%`	Max efficiency of Class B amplifier

Module 2 — Operational Amplifiers

Circuit	Formula	Notes
Inverting Amplifier Output	`Vout = −(Rf/Rin) × Vin`	Negative = 180° phase inversion
Inverting Amplifier Gain	`Av = −Rf / Rin`	Negative gain value
Non-Inverting Amplifier Gain	`Av = 1 + (Rf / R1)`	Always ≥ 1, in phase
Summing Amplifier Output	`Vout = −Rf(V1/R1 + V2/R2 + V3/R3)`	If R1=R2=R3=R: Vout=−(Rf/R)(V1+V2+V3)
Integrator Output	`Vout = −(1/RC) ∫ Vin dt`	Square input → triangular output
Differentiator Output	`Vout = −RC × dVin/dt`	Triangular input → square output
Comparator Rule	`If V+ > V−: Vout = +Vsat` `If V+ < V−: Vout = −Vsat`	Open-loop op-amp
Virtual Ground	`V− ≈ 0V (inverting amp with −ve feedback)`	Key analysis tool

Module 3 — Number Systems & Boolean

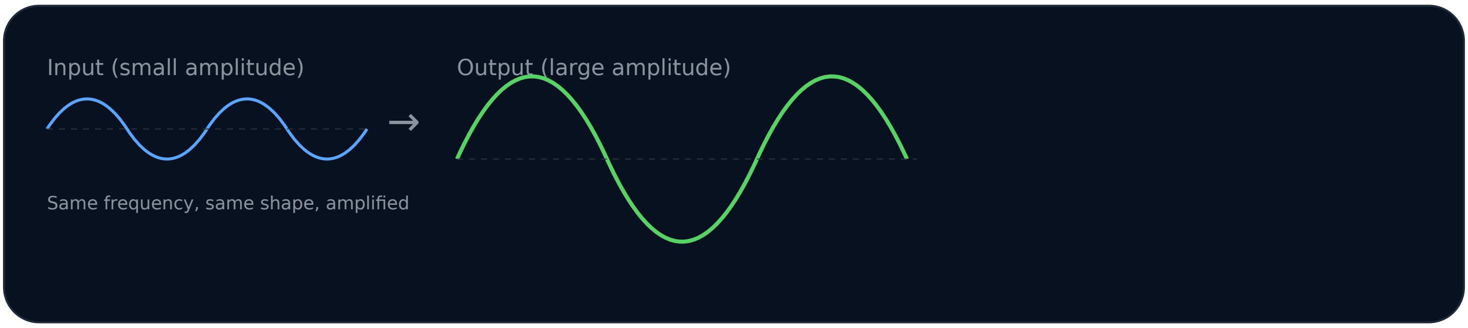
Topic	Formula / Rule
Binary to Decimal	$N = b_n \times 2^n + ... + b_1 \times 2^1 + b_0 \times 2^0$
Hex to Decimal	$N = h_n \times 16^n + ... + h_1 \times 16^1 + h_0 \times 16^0$
Two's Complement	`Two's complement = One's complement + 1`
De Morgan 1	`(A·B)' = A' + B'`
De Morgan 2	`(A+B)' = A'·B'`
XOR	`A⊕B = A'B + AB'`
Boolean Identity	`A+0=A, A·1=A, A+1=1, A·0=0`
Boolean Complement	`A+A'=1, A·A'=0, (A')'=A`

Module 4 — Combinational & Sequential

Circuit	Formula
Half Adder Sum	`S = A ⊕ B`
Half Adder Carry	`C = A · B`
Half Subtractor Difference	`D = A ⊕ B`
Half Subtractor Borrow	`Br = A'B`
Full Adder Sum	`S = A ⊕ B ⊕ Cin`
Full Adder Carry	`Cout = AB + BCin + ACin`
MOD-N Counter	N flip-flops → MOD-2^N maximum
JK FF Toggle	`Q(next) = Q' when J=K=1`
D FF	`Q(next) = D`
T FF	`Q(next) = Q' when T=1; Q when T=0`

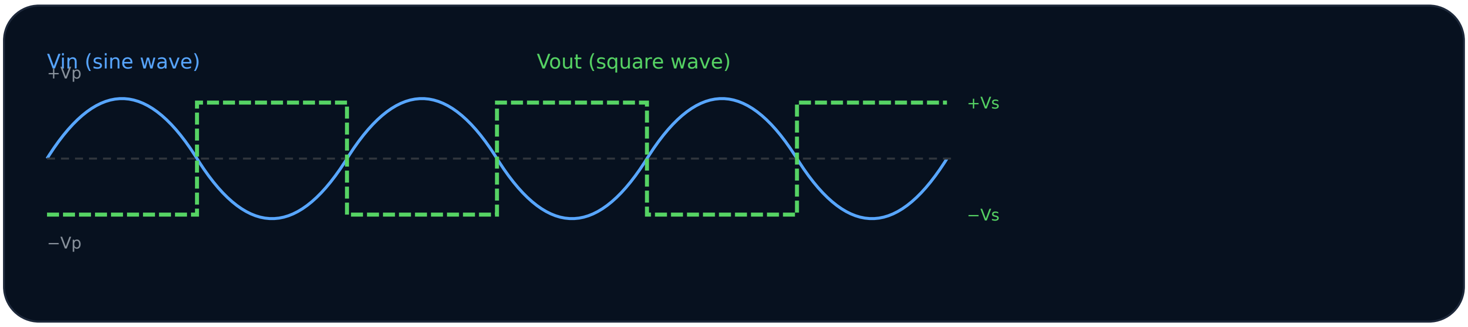
〰️ Waveform Bank

Amplifier Input-Output

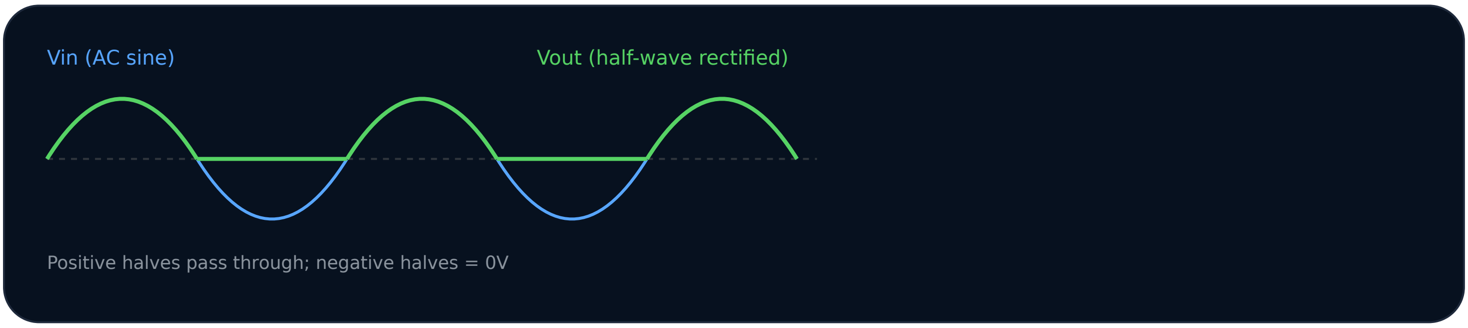


Astable Multivibrator Output (Square Wave)

Op-Amp Comparator / Zero Crossing Detector

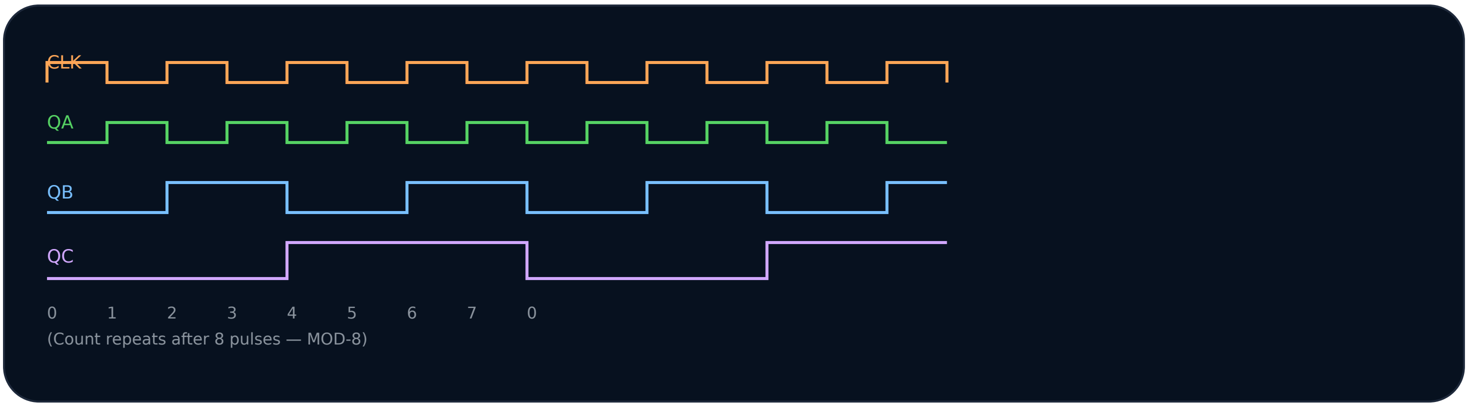


Precision Rectifier Waveform

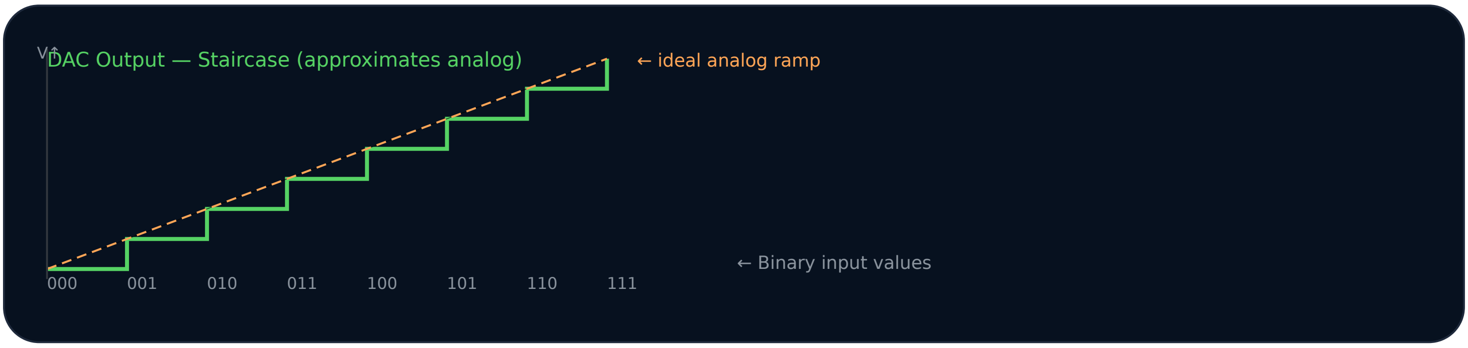


Clock Pulse & Flip-Flop Timing

MOD-8 Ripple Counter Timing



DAC Staircase Output



📐 K-Map Solving Bank

Purpose of K-Map

K-Map (Karnaugh Map) provides a systematic, visual method to minimise Boolean expressions. It avoids lengthy algebraic manipulation. The map uses Gray code ordering to ensure adjacent cells differ by exactly one variable — allowing easy identification of groupings that simplify the expression.

> 🔖 K-Map: Boolean expression □□□ chart-□ write □□□□□□ visually simplify □□□□□□□□□ method. Adjacent cells 1 bit differ → group □□□□□□ simplify □□□□□□□. Algebra use □□□□□□□□□.

Step-by-Step SOP Simplification Using K-Map

1. Write the Boolean function as a minterm list: Y = Σm(…)
2. Draw the K-Map with correct dimensions (2-var: 2×2, 3-var: 2×4). Use Gray code ordering for columns.
3. Fill 1s in all cells corresponding to minterm numbers. Fill 0s in remaining cells.
4. Identify groups of 1s. Group sizes must be powers of 2 (1,2,4,8). Make groups as large as possible. Groups can wrap around edges.

5. For each group: identify which variable(s) remain constant. Those constants form the product term.
6. OR (sum) all product terms → final simplified SOP expression.

Solved K-Map Example — 3 Variables, Quad

🗺️ **Simplify: Y = Σm(0, 2, 4, 6)**

Step 1: All minterms have C=0. Fill K-Map:

A\BC	00	01	11	10
0	**1(m0)**	0(m1)	0(m3)	**1(m2)**
1	**1(m4)**	0(m5)	0(m7)	**1(m6)**

Step 2: Group: m0,m2,m4,m6 — all 4 cells (quad). Columns 00 and 10 (C=0 in both). A varies (0,1). B varies (0,1). Only constant: C=0.

Simplified SOP: **Y = C'**

Solved K-Map Example — 3 Variables, Two Groups

🗺️ **Simplify: Y = Σm(1, 3, 5, 7)**

A\BC	00	01	11	10
0	0(m0)	1(m1)	1(m3)	0(m2)
1	0(m4)	1(m5)	1(m7)	0(m6)

Group (blue quad): m1,m3,m5,m7 — columns 01 and 11 → C=1 constant. A varies. B varies. → **C**

Simplified SOP: **Y = C**

Solved K-Map Example — Multiple Groups

🗺️ **Simplify: Y = Σm(0, 1, 4, 5, 6)**

A\BC	00	01	11	10
0	1(m0)	1(m1)	0(m3)	0(m2)
1	1(m4)	1(m5)	0(m7)	1(m6)

Group 1 (orange quad): m0,m1,m4,m5 → BC=00,01 columns; A=0,1; B=0,0; C=0,1 → B=0 constant, C varies, A varies → **B'**

Group 2 (blue pair): m4,m6 → A=1; BC=00,10 → A=1 constant, C=0 constant, B varies → **AC'**

Simplified SOP: **Y = B' + AC'**

Check: m6 is covered by AC'. All 1s covered. ✓

❓ Expected Questions — Module-wise

Short Answer (2 marks)

1. Define amplifier. State the formula for voltage gain.
2. What is Barkhausen's criterion? State the two conditions.
3. Name four types of coupling used in multistage amplifiers.
4. What is the difference between Class A and Class B amplifier?
5. Define oscillator. How does it differ from an amplifier?
6. What is positive feedback? Where is it used?
7. Name two types of multivibrators and their stable states.
8. State two applications of crystal oscillator.

Paragraph (5 marks)

1. Explain RC phase shift oscillator with a circuit diagram and working.
2. Draw and explain the complementary symmetry push-pull amplifier (Class B).
3. Explain positive feedback and negative feedback with block diagrams.
4. Compare Class A, Class B, and Class C amplifiers in a table.
5. Describe astable multivibrator operation with a circuit diagram.

Essay / Long Answer (10 marks)

1. Explain the working of an RC coupled two-stage transistor amplifier with a block diagram. State the advantages of RC coupling.
2. What is an oscillator? State Barkhausen's criterion. Explain the working of an RC phase shift oscillator with circuit diagram and waveform.
3. Classify power amplifiers. Compare Class A, B, and C amplifiers. Explain the working of a Class B complementary symmetry push-pull amplifier.

Short Answer (2 marks)

1. What is virtual ground? Where is it applied?
2. List four parameters of an ideal op-amp.
3. What is CMRR? State the ideal value.
4. State the gain formula for inverting and non-inverting amplifiers.
5. What is slew rate?
6. What is a zero crossing detector?
7. Why is a precision rectifier better than a normal rectifier?
8. What is the output of a differentiator for a triangular wave input?

Paragraph (5 marks)

1. Derive the output voltage expression of an inverting amplifier using virtual ground concept.
2. Explain the working of a summing amplifier with circuit diagram and output expression.
3. Explain the op-amp integrator circuit. Derive its output equation.
4. Explain the half-wave precision rectifier with circuit diagram and input-output waveforms.
5. Explain positive voltage level detector and zero crossing detector with diagrams.

Essay (10 marks)

1. Draw the op-amp symbol. Compare ideal and practical op-amp parameters in a table. Explain virtual ground concept.
2. With circuit diagram, derive the output of an inverting amplifier. Also explain non-inverting amplifier and its gain. Compare the two.
3. Explain the op-amp differentiator and integrator circuits with derivations, waveforms, and applications.

Short Answer (2 marks)

1. Convert $(1010\ 1100)_2$ to decimal.
2. State De Morgan's two theorems.
3. Find two's complement of $(10110)_2$.
4. Draw symbol and truth table of XOR gate.
5. Why are NAND and NOR called universal gates?
6. Convert $(47)_{10}$ to binary and hexadecimal.
7. What is BCD? Convert $(36)_{10}$ to BCD.
8. What is the difference between SOP and POS forms?

Paragraph (5 marks)

1. Perform binary subtraction: $(1100)_2 - (0101)_2$ using two's complement method.
2. Simplify $Y = \Sigma m(0,1,2,3,5)$ using 3-variable K-Map. Draw the K-Map and show groupings.
3. Simplify using Boolean algebra: $Y = AB + A(B+C) + B(B+C)$.
4. Explain NAND and NOR gates as universal gates. Show how to implement NOT, AND, OR using NAND gates only.
5. Explain K-Map grouping rules with examples for 2-variable and 3-variable maps.

Essay (10 marks)

1. Explain binary, decimal, and hexadecimal number systems. Perform conversions: $(245)_{10}$ to binary and hex; $(FF)_{16}$ to decimal; $(11001101)_2$ to decimal.
2. Explain basic logic gates (AND, OR, NOT, NAND, NOR, XOR) with symbols and truth tables. State De Morgan's theorems with proof using truth tables.

3. Simplify the following using K-Map: $Y = \Sigma m(0,1,3,4,5,7)$. Draw K-Map, show all groups, write simplified SOP.

Short Answer (2 marks)

1. Write the Boolean expressions for half adder Sum and Carry.
2. Draw the truth table of a JK flip-flop.
3. What is the difference between synchronous and asynchronous counters?
4. List four types of shift registers.
5. What is MOD-8 counter? How many flip-flops are required?
6. Define combinational circuit and sequential circuit.
7. What is a DAC? Name two types.
8. State the function of a T flip-flop when T=1.

Paragraph (5 marks)

1. Explain full adder using two half adders. Draw logic diagram and write truth table.
2. Explain SISO and SIPO shift registers using negative edge triggered D flip-flops. Draw logic diagrams.
3. Explain MOD-8 ripple up counter with circuit diagram and count sequence table.
4. Explain 4×1 multiplexer with truth table. How is it different from a demultiplexer?
5. Explain R-2R ladder DAC with a diagram. State its advantage over binary weighted DAC.

Essay (10 marks)

1. Explain SR, D, JK, and T flip-flops with symbols, truth tables, and operation. State applications of flip-flops.
2. Explain MOD-8 ripple up counter, MOD-7, and MOD-6 counters using JK flip-flops. Draw diagrams and count sequence tables.
3. Explain the concept of DAC and ADC. Draw and explain R-2R ladder DAC and ramp type ADC with block diagrams.

📝 Practice Problems

Number Systems & Arithmetic

1. Convert $(198)_{10}$ to binary and hexadecimal.
2. Convert $(1011\ 0110)_2$ to decimal and hexadecimal.
3. Add: $(11011)_2 + (10110)_2$. Verify by converting to decimal.
4. Subtract $(1001)_2$ from $(1101)_2$ using two's complement method.

5. Multiply: $(1011)_2 \times (101)_2$.
6. Convert $(3A7)_{16}$ to decimal.
7. Find BCD of $(94)_{10}$.
8. Find two's complement of $(110011)_2$.

Boolean Algebra

1. Simplify: Y = AB + A'B + AB'
2. Apply De Morgan's: (A'B + C)'
3. Prove: A + A'B = A + B using Boolean algebra.
4. Simplify: Y = (A + B)(A + C)
5. Convert Y = AB' + A'B to NAND-NAND form.

K-Map Simplification

1. Simplify using K-Map: Y = Σm(1, 2, 3, 7) (3-variable)
2. Simplify using K-Map: Y = Σm(0, 2, 4, 6) (3-variable)
3. Simplify using K-Map: Y = Σm(0, 1, 4, 5, 6, 7) (3-variable)
4. Simplify: Y = Σm(3, 5, 6, 7) (3-variable)
5. Simplify using 2-variable K-Map: Y = Σm(0, 1, 2)

Op-Amp Problems

1. An inverting amplifier has Rin = 10kΩ, Rf = 100kΩ. Find Av and Vout if Vin = 0.5V.
2. A non-inverting amplifier has R1 = 5kΩ, Rf = 45kΩ. Find Av and Vout for Vin = 1V.
3. A summing amplifier has R1=R2=R3=10kΩ, Rf=30kΩ. V1=1V, V2=2V, V3=3V. Find Vout.
4. For an integrator with R=10kΩ, C=1μF, Vin=2V (DC step). Write the output expression.

Counters and Flip-flops

1. Draw the state table for a MOD-5 counter using JK flip-flops.
2. A 4-bit SIPO shift register initially contains 0000. Serial input = 1011 (MSB first). Show the state after each clock pulse.
3. Design a MOD-6 ripple counter and draw its state table.

4. For a D flip-flop, if D=1 and Q=0, what is Q after a clock pulse?

⚡ Quick Revision

Read this the night before the exam. All key points in one place.

Module 1 — Amplifiers & Oscillators

- Av = Vout/Vin · Ai = Iout/Iin · Ap = Av×Ai
- RC coupling: capacitor passes AC, blocks DC
- Class A: full cycle, ~30% eff, least distortion
- Class B: half cycle, ~78.5% eff, crossover distortion
- Class C: <180°, highest eff, most distortion, RF use
- Push-pull: NPN+PNP, full cycle output, 78.5%
- +ve feedback → gain increases → oscillations
- −ve feedback → gain reduces → stability
- Barkhausen: |Aβ|=1 AND phase shift=0° or 360°
- RC phase shift: f = 1/(2π√6 RC), needs Av≥29
- Crystal oscillator: high Q, very stable frequency
- Astable: no stable state → continuous square wave
- Bistable: 2 stable states = flip-flop

Module 2 — Op-Amps

- Ideal: Gain=∞, Zin=∞, Zout=0, BW=∞
- IC 741: Gain≈200k, Zin≈2MΩ, Zout≈75Ω
- Virtual ground: V− ≈ 0V in inverting amp
- Inverting: Vout = −(Rf/Rin)Vin, gain=−Rf/Rin
- Non-inverting: Av = 1 + Rf/R1, in phase
- Summing: Vout = −Rf(V1/R1+V2/R2+...)
- Integrator: C in feedback → Vout=−(1/RC)∫Vin dt
- Differentiator: C at input → Vout=−RC(dVin/dt)
- Comparator: V+>V− → +Vsat; V+<V− → −Vsat
- Zero crossing: sine → square; Vref=0V
- Precision rectifier: removes 0.7V diode drop problem

Module 3 — Digital Basics & K-Map

- Binary: base 2 (0,1). Hex: base 16 (0-9,A-F)
- Binary→Dec: positional weights (2^n)
- Dec→Binary: repeated ÷2, read remainders upward

Module 4 — Sequential & Data Conversion

- Half adder: S=A⊕B, C=AB
- Half subtractor: D=A⊕B, Br=A'B
- Full adder: S=A⊕B⊕Cin, Cout=AB+BCin+ACin

- Two's complement = one's complement + 1
- BCD: each decimal digit = 4 bits
- AND: output 1 only when ALL inputs 1
- OR: output 1 when ANY input 1
- NAND, NOR: universal gates
- XOR: output 1 when inputs DIFFERENT (A⊕B)
- De Morgan 1: (AB)' = A'+B'
- De Morgan 2: (A+B)' = A'·B'
- K-Map: groups must be powers of 2 (1,2,4,8)
- K-Map columns: Gray code (00,01,11,10)
- Larger group = fewer literals = simpler expression

- MUX: many inputs → 1 output (data selector)
- DEMUX: 1 input → many outputs
- SR FF: S=set, R=reset; S=R=1 forbidden
- D FF: Q(next) = D (no forbidden state)
- JK FF: J=K=1 → toggle; no forbidden state
- T FF: T=1 → toggle; T=0 → hold
- SISO/SIPO/PIPO/PISO: 4 shift register types
- MOD-8 counter: 3 JK FFs, counts 0→7
- MOD-7: NAND reset when count=7
- MOD-6: NAND reset when QB=QC=1 (count=6)
- Binary weighted DAC: R,2R,4R,8R... (impractical)
- R-2R ladder DAC: only R and 2R (practical)
- Ramp ADC: counter+DAC+comparator

One-Liners to Memorise

🔑	Barkhausen = $\lvert A\beta \rvert=1$ AND phase=0°/360°
🔑	Inverting amp gain = $-R_f/R_{in}$ (negative = inverted)
🔑	Non-inverting gain = $1 + R_f/R_1$ (always ≥1)
🔑	Integrator output = $V_{out} = -(1/RC)\int V_{in}\, dt$
🔑	K-Map columns always in Gray code: 00→01→11→10
🔑	Two's complement = one's comp + 1
🔑	Half adder Sum = XOR, Carry = AND
🔑	JK FF: J=K=1 → toggle (Q flips each clock)
🔑	MOD-8 = 3 FFs, counts 8 states (0–7)
🔑	R-2R DAC: only 2 resistor values, high precision
🔑	Virtual ground: $V- \approx 0V$ in inverting op-amp
🔑	NAND and NOR = universal gates

📄 Model Question Paper — with Full Answers

SITTTR · Diploma in EEE · Semester 3

3031 — Analog & Digital Circuits

Time: 3 Hours | Maximum Marks: 100 | All questions carry equal marks in each part

Answer ALL questions. Draw circuit diagrams wherever required. Write neatly.

PART A — Short Answer (10 × 3 = 30 marks)

Answer ALL ten questions. Each question carries 3 marks.

Q1. State Barkhausen's criterion for sustained oscillations.

Barkhausen's Criterion: For an amplifier with positive feedback to produce sustained oscillations, two conditions must be simultaneously satisfied:

1. **Loop Gain Condition:** $|A\beta| = 1$ (magnitude of loop gain must be exactly 1). If $|A\beta| < 1$, oscillations die out; if $|A\beta| > 1$, oscillations grow unstably.
2. **Phase Condition:** Total phase shift around the loop = 0° or 360°. This ensures positive feedback (output fed back in phase with input).

Where A = open-loop gain of amplifier, β = feedback factor.

Q2. Compare Class A and Class B power amplifiers.

Parameter	Class A	Class B
Conduction angle	360°	180°
Efficiency	~25–30%	~78.5%
Distortion	Very low	Crossover distortion
Q-point	Centre of load line	At cut-off
Application	Pre-amplifiers	Power output stages

Q3. What is virtual ground? Why is it important in op-amp analysis?

Virtual Ground: In an inverting amplifier with negative feedback, the inverting input terminal (V−) is held at approximately 0V potential, even though it is not physically connected to ground. This is because the very high open-loop gain of the op-amp forces

V+ ≈ V− through feedback action. Since V+ = 0V (grounded), V− ≈ 0V — virtual ground.

Importance: Virtual ground simplifies circuit analysis. We can assume V− = 0V and use Kirchhoff's current law to derive output voltage expressions easily without complex nodal analysis.

Q4. Convert $(11001010)_2$ to decimal and hexadecimal.

Binary to Decimal:
$= 1\times2^7 + 1\times2^6 + 0\times2^5 + 0\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0$
$= 128 + 64 + 0 + 0 + 8 + 0 + 2 + 0 =$ **$(202)_{10}$**

Decimal to Hex:
202 ÷ 16 = 12 remainder 10 = A (LSB)
12 ÷ 16 = 0 remainder 12 = C (MSB)
Hex = **$(CA)_{16}$**

Q5. State De Morgan's theorems with Boolean expressions.

De Morgan's First Theorem: (A · B)' = A' + B'
The complement of a product (AND) equals the sum (OR) of the individual complements.

De Morgan's Second Theorem: (A + B)' = A' · B'
The complement of a sum (OR) equals the product (AND) of the individual complements.

These theorems are used to simplify Boolean expressions, convert NAND ↔ NOR, and implement logic using universal gates.

Q6. Write the truth table of a JK flip-flop.

J	K	Q(present)	Q(next)	Operation
0	0	Q	Q	Hold (no change)
0	1	X	0	Reset
1	0	X	1	Set
1	1	Q	Q'	Toggle (complement)

The JK flip-flop eliminates the forbidden state of SR flip-flop. When J=K=1, Q toggles on each clock pulse.

Q7. What is K-Map? State its advantages.

K-Map (Karnaugh Map): A graphical tool for minimising Boolean expressions. It arranges minterm values in a 2D grid where adjacent cells differ by only one variable (Gray code order). By grouping adjacent 1s in powers of 2, simplified product terms are directly obtained.

Advantages:

- Faster and more systematic than algebraic simplification
- Minimises both the number of terms and literals
- Visual — groupings are easy to identify
- Reduces hardware (fewer gates needed)

Q8. Classify shift registers.

Shift registers are classified based on the method of data input and output:

Type	Full Form	Application
SISO	Serial In Serial Out	Time delay circuits
SIPO	Serial In Parallel Out	Serial-to-parallel conversion
PIPO	Parallel In Parallel Out	Data storage/transfer
PISO	Parallel In Serial Out	Parallel-to-serial conversion

Q9. What is a multiplexer? What is the function of select lines in a 4×1 MUX?

Multiplexer (MUX): A combinational circuit that selects one of several input data lines and routes it to a single output line based on binary select inputs. It acts as a many-to-one data switch.

Select lines in 4×1 MUX: A 4×1 MUX has 4 data inputs (I0–I3), 2 select lines (S1, S0), and 1 output Y. The 2-bit binary value on select lines determines which input is passed to the output:

- S1S0 = 00 → Y = I0
- S1S0 = 01 → Y = I1
- S1S0 = 10 → Y = I2
- S1S0 = 11 → Y = I3

Q10. Why is R-2R ladder DAC preferred over binary weighted DAC?

Binary Weighted DAC: Requires resistors of values R, 2R, 4R, 8R, 16R... for each additional bit. For large bit numbers, very high resistance values become impractical and difficult to fabricate with precision in IC form.

R-2R Ladder DAC: Uses only two resistor values — R and 2R — regardless of the number of bits. The R-2R ladder network provides binary-weighted voltages through a voltage divider principle without needing many different resistor values.

Reasons for preference: Easier IC fabrication with high accuracy, better matching of resistors, lower cost, and practical for any bit width.

PART B — Short Essay (5 × 8 = 40 marks)

Answer any FIVE questions. Each carries 8 marks.

Q11. Explain the working of an RC phase shift oscillator. Draw its circuit diagram and state the frequency of oscillation.

RC Phase Shift Oscillator:

Circuit: Consists of a transistor CE amplifier and a feedback network made of three RC sections in series.

Working:

1. The transistor CE amplifier provides a phase shift of 180° (input-to-output inversion).
2. The RC feedback network (3 RC sections) provides an additional phase shift of 180°.
3. Each RC section contributes 60°, total = 3 × 60° = 180°.
4. Total loop phase shift = 180° (transistor) + 180° (RC network) = 360°.
5. This satisfies Barkhausen phase condition (0° or 360°).
6. The transistor gain is set to satisfy |Aβ| = 1 (Av ≥ 29).
7. The circuit self-starts due to noise and sustains at the oscillation frequency.

Frequency of Oscillation:

```
f = 1 / (2π√6 RC)
```

Applications: Audio frequency generation, function generators, signal sources in testing equipment.

Q12. Derive the output voltage expression of an inverting op-amp amplifier. Compare it with a non-inverting amplifier.

Inverting Amplifier Derivation:
Circuit: Vin applied to inverting input (−) through Rin. Feedback resistor Rf from output to inverting input. Non-inverting input (+) at ground.

1. Apply virtual ground: V− = 0V (since V+ = 0V and high gain forces V− ≈ V+).
2. Current through Rin: I_1 = (Vin − 0)/Rin = Vin/Rin
3. Op-amp draws no input current → all I_1 flows through Rf.
4. Voltage at output: Vout = 0 − I_1 × Rf = −(Vin/Rin) × Rf

```
Vout = −(Rf/Rin) × Vin     Av = −Rf/Rin
```

The negative sign indicates 180° phase inversion.

Comparison:

Parameter	Inverting	Non-Inverting
Input terminal	Inverting (−)	Non-inverting (+)
Phase	180° shifted	In phase (0°)
Gain formula	Av = −Rf/Rin	Av = 1 + Rf/R1
Min gain	Can be <1	Always ≥ 1
Input impedance	= Rin	Very high (Zin of op-amp)

Q13. Simplify using K-Map: Y = Σm(0, 1, 2, 3, 4, 6). Draw K-Map, identify groups, and write simplified SOP.

3-Variable K-Map — Variables: A, B, C
Fill 1s at m0,m1,m2,m3,m4,m6. Fill 0s at m5,m7.

A\BC	00	01	11	10
0	1(m0)	1(m1)	1(m3)	1(m2)
1	1(m4)	0(m5)	0(m7)	1(m6)

Groups:

- **Group 1 (orange, octet-like quad):** m0,m1,m2,m3 → A=0 row — A=0 constant, B varies, C varies → term = **A'**

- **Group 2 (blue, pair):** m4,m6 → A=1, BC=00,10 → C=0 constant, A=1 constant, B varies → term = **AC'**

```
Y = A' + AC'
```

Can further simplify using absorption: Y = A' + C' (by consensus theorem: A'+AC' = A'+C').
Final simplified SOP: Y = A' + C'

> Q14. Explain the working of a MOD-8 ripple counter. Draw the logic diagram and count sequence table.

MOD-8 Ripple Counter:

Uses 3 negative-edge-triggered JK flip-flops (FF-A, FF-B, FF-C) with J=K=1 (toggle mode) in all flip-flops.

Working:

- FF-A (LSB) is clocked by the external clock input. Toggles on every negative edge.
- FF-B is clocked by QA output. Toggles when QA changes from 1→0 (negative edge of QA).
- FF-C (MSB) is clocked by QB output. Toggles when QB changes 1→0.
- Outputs QC,QB,QA form the 3-bit binary count.
- After 8 clock pulses, the count returns to 000 — completing one full cycle.

Count Sequence (0 to 7):

CLK Pulse	QC	QB	QA	Decimal
0	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7
8 (reset)	0	0	0	0 ← repeats

Note: QA frequency = CLK/2, QB = CLK/4, QC = CLK/8. Each FF divides frequency by 2.

> Q15. Explain the operation of a Full Adder using two Half Adders. Draw the logic diagram and write the truth table.

Full Adder: Adds three bits — A, B, and Carry-in (Cin) — producing Sum (S) and Carry-out (Cout).

Construction using 2 Half Adders:

- **Half Adder 1 (HA1):** Adds A and B → S1 = A⊕B, C1 = AB
- **Half Adder 2 (HA2):** Adds S1 and Cin → S = S1⊕Cin = A⊕B⊕Cin, C2 = S1·Cin
- **OR gate:** Cout = C1 + C2 = AB + (A⊕B)·Cin

Boolean Expressions:

```
Sum S = A ⊕ B ⊕ Cin
```

```
Cout = AB + BCin + ACin
```

Truth Table:

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

PART C — Essay (2 × 15 = 30 marks)

Answer any TWO questions. Each carries 15 marks.

Q16. (a) Explain op-amp integrator and differentiator circuits with derivations and waveforms. (b) Explain half-wave precision rectifier with circuit and waveforms. (7+8 = 15)

(a) Op-Amp Integrator:
Circuit: Rin at input, capacitor C in feedback path.
Derivation: Virtual ground → V− = 0. Current through R: I = Vin/R. This current charges C. Voltage across C: Vc = (1/C)∫I·dt. Output: Vout = −Vc.

```
Vout = −(1/RC) ∫ Vin dt
```

Waveforms: Square wave input → triangular wave output (capacitor charges/discharges linearly).

Op-Amp Differentiator:
Circuit: Capacitor C at input, Rf in feedback path.
Derivation: Current through C: I = C·d(Vin)/dt. This flows through Rf. Vout = −I·Rf.

```
Vout = −RC · dVin/dt
```

Waveforms: Triangular wave input → square wave output (derivative of a ramp = constant).

(b) Half-Wave Precision Rectifier:
Problem: Normal diode has 0.7V drop — cannot rectify signals <0.7V.
Solution: Place diode D1 inside the feedback loop of an op-amp.
Working:

- Positive half of Vin: Op-amp output goes positive → D1 conducts → feedback establishes → Vout = Vin (no diode drop since op-amp compensates)
- Negative half of Vin: Op-amp output goes to −Vsat → D1 reverse biased → no feedback → Vout = 0V

Output: Positive half cycles pass accurately (even millivolt signals), negative halves are clipped at 0V. Result = precise half-wave rectified output.

Q17. (a) Explain all four flip-flops (SR, D, JK, T) with truth tables. (b) Explain SISO and SIPO shift registers with logic diagrams. (8+7 = 15)

(a) Flip-Flops:

SR Flip-Flop: Set-Reset FF. S=1 → Q=1 (Set). R=1 → Q=0 (Reset). S=R=0 → Hold. S=R=1 → Forbidden (indeterminate state). Used in latches.

D Flip-Flop: Data FF. Q(next) = D. On clock edge, Q follows D. Eliminates forbidden state. Used in registers and memory. Simple, predictable operation.

JK Flip-Flop: Universal FF. J=K=0 → Hold. J=1,K=0 → Set (Q=1). J=0,K=1 → Reset (Q=0). J=K=1 → Toggle (Q complements). No forbidden state. Most versatile flip-flop.

T Flip-Flop: Toggle FF. T=0 → Hold (Q unchanged). T=1 → Toggle (Q = Q'). Derived from JK by connecting J=K=T. Used in binary counters (frequency dividers).

Truth Tables: [Refer to Module 4 section above for individual tables]

(b) SISO Shift Register:
Serial In, Serial Out. 4 D flip-flops in series. Data enters one bit at a time at the first FF and exits one bit at a time from the last FF. Takes 4 clock pulses to shift 4-bit data through.
Use: Time delay circuits — data appears at output delayed by N clock pulses (N = number of FFs).

SIPO Shift Register:
Serial In, Parallel Out. Same series connection of 4 D FFs. Data enters serially. After 4 clock pulses, all 4 bits are available simultaneously at Q outputs of all 4 FFs.
Use: Serial-to-parallel data conversion. Used in communication receivers to convert serial data from a channel into parallel bytes for a microprocessor.

Both use negative-edge-triggered D flip-flops. Each FF connected: D of next FF = Q of current FF. Common clock applied to all FFs.

> Q18. (a) Explain number systems (Binary, Decimal, Hex) and number conversions. Perform: (i) $(156)_{10}$ to binary (ii) $(1010\ 1111)_2$ to decimal (iii) $(BC)_{16}$ to decimal. (b) Explain binary arithmetic: addition, subtraction using two's complement. (8+7 = 15)

(a) Number Systems:
Decimal (Base 10): Uses digits 0–9. Position weights are powers of 10. Used in everyday calculations.
Binary (Base 2): Uses only 0 and 1. Position weights = powers of 2 (1,2,4,8,16...). Fundamental language of digital systems.
Hexadecimal (Base 16): Uses 0–9 and A–F (A=10,B=11,...,F=15). Each hex digit = 4 bits. Compact representation of binary numbers.

(i) $(156)_{10}$ to Binary:
156÷2=78 r **0**; 78÷2=39 r **0**; 39÷2=19 r **1**; 19÷2=9 r **1**; 9÷2=4 r **1**; 4÷2=2 r **0**; 2÷2=1 r **0**; 1÷2=0 r **1**
Read upward: **$(10011100)_2$**

(ii) $(10101111)_2$ to Decimal:
= 1×128+0×64+1×32+0×16+1×8+1×4+1×2+1×1 = 128+32+8+4+2+1 = **$(175)_{10}$**

(iii) $(BC)_{16}$ to Decimal:
B=11, C=12. = $11\times16^1 + 12\times16^0$ = 176+12 = **$(188)_{10}$**

(b) Binary Addition:
Rules: 0+0=0; 0+1=1; 1+0=1; 1+1=10 (0, carry 1).
Example: $(1011)_2 + (0110)_2$:

```
  1011
+ 0110
------
 10001 = (17)_10 ✓ (11+6=17)
```

Binary Subtraction Using Two's Complement:

Method: Instead of direct subtraction, add the two's complement of the subtrahend.

Example: $(1010)_2 - (0011)_2$:

1. Two's complement of 0011: One's comp = 1100; + 1 = 1101
2. Add: 1010 + 1101 = 10111
3. Discard carry: 0111 = $(7)_{10}$ ✓ (10−3=7)

📚 References and Online Resources

Textbooks

- **Boylestad R.L. & Nashelsky L.** — Electronic Devices and Circuit Theory (Pearson)
- **Millman J. & Halkias C.** — Electronics (Tata McGraw-Hill)
- **Floyd T.L.** — Digital Fundamentals (Pearson)
- **Mano M.M.** — Digital Design (Pearson)
- **Gayakwad R.A.** — Op-Amps and Linear Integrated Circuits (PHI)

Online Resources

- SITTTR Kerala Official Site — Model QPs, syllabus documents
- — Amplifiers, op-amps, digital circuits
- — Freenoreferrer">NPTEL — Free video lectures on analog and digital electronics
-